# AI-Powered Oral Cancer Diagnosis: Automated Report Generation for Enhanced Healthcare

**CAPSTONE PROJECT REPORT**

**Submitted by**

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**in partial fulfillment for the award of the degree**

**of**

**BACHELOR OF TECHNOLOGY**

**IN**

**COMPUTER SCIENCE AND ENGINEERING**

**SCHOOL OF COMPUTING**

**COMPUTER SCIENCE AND ENGINEERING**

**KALASALINGAM ACADEMY OF RESEARCH AND EDUCATION**

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# DECLARATION

We affirm that the project work titled **"AI-Powered Oral Cancer Diagnosis: Automated Report Generation for Enhanced Healthcare"** being submitted in partial fulfillment for the award of the degree of **Bachelor of Technology in Computer Science and Engineering** is the original work carried out by us. It has not formed part of any other project work submitted for the award of any degree or diploma, either in this or any other University.

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This is to certify that the above statement made by the candidate is correct to the best of my knowledge.

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Under sec. 3 of UGC Act 1956. Accredited by NAAC with "A++" Grade

# BONAFIDE CERTIFICATE

Certified that this project report **"AI-Powered Oral Cancer Diagnosis: Automated Report Generation for Enhanced Healthcare"** is the Bonafide work of "**VENNAPUSA GIRIVARDHAN REDDY (99220040360), PRALAPALLI BALA BHARGAV (99220040329), PENCHINENI SASI KUMAR (99220040332), PAPANI SIVA VENKATA SAI (99220040328)"** who carried out the project work under my supervision.

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We are deeply grateful to the late **"Kalvivallal" Thiru T. Kalasalingam,** Chairman of the Kalasalingam Group of Institutions, and to "**Illayavallal" Dr. K. Sridharan**, Chancellor, as well as **Dr. S. Shasi Anand**, Vice President, who has been a guiding light in all our university's endeavours.

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**SCHOOL OF COMPUTING**

**COMPUTER SCIENCE AND ENGINEERING**

**PROJECT SUMMARY**

| | | |
|---|---|---|
| Project Title | Real-Time Pose Estimation and Joint Analysis with MediaPipe and OpenCV | |
| Project Team Members (Name with Register No) | VENNAPUSA GIRIVARDHAN REDDY (99220040360), PRALAPALLI BALA BHARGAV (99220040329), PENCHINENI SASI KUMAR (99220040332), PAPANI SIVA VENKATA SAI (99220040328) | |
| Guide Name/Designation | Marimuthu. T | |
| Program Concentration Area | Artificial Intelligence and Computer Vision (Intelligent Systems) | |
| Technical Requirements | Machine Learning, Deep Learning, Computer Vision, Digital Image Processing, Python Programming, PyTorch, EfficientNetV2, ConvNeXt, OpenCV, Flask, Grad-CAM, Image Classification, Explainable AI, Email Integration, NumPy, PIL , Torchvision, Image Preprocessing, Real-Time Inference, Web Application Development | |
| Engineering standards and realistic constraints in these areas | | |
| **Area** | **Codes & Standards / Realistic Constraints** | **Tick ✓** |
| Economic | The system is designed using open-source frameworks such as MediaPipe and OpenCV, minimizing licensing costs. It operates on standard consumer hardware, reducing deployment expenses and ensuring affordability for educational institutions and rehabilitation centers. | ✓ |
| Environmental | The system is fully software-based and does not require specialized hardware, thereby reducing electronic waste. It minimizes power consumption by using lightweight models and optimized real-time processing. | ✓ |
| Social | The framework promotes social well-being by supporting fitness monitoring, posture correction, and rehabilitation. It enhances accessibility by enabling low-cost motion analysis tools for home-based users. | ✓ |

| | | |
|---|---|---|
| Ethical | The system adheres to ethical AI principles by ensuring data privacy, avoiding facial recognition storage, and focusing only on skeletal landmark data. No personally identifiable information (PII) is stored. | ✓ |
| Health and Safety | The system contributes to injury prevention by identifying incorrect postures and joint misalignment. It reduces the risk of musculoskeletal disorders through real-time posture feedback | ✓ |
| Manufacturability | Since the project is software-oriented, deployment only requires standard computing devices and webcams. The modular design allows easy integration into mobile or embedded systems. | ✓ |
| Sustainability | The lightweight computational model ensures lower energy consumption. The system encourages long-term use in healthcare and fitness applications, supporting sustainable digital health solutions. | ✓ |

## REALISTIC CONSTRAINTS

**Economic Constraints:**

The proposed pose estimation system is economically feasible as it relies entirely on open-source technologies including MediaPipe and OpenCV. These frameworks eliminate licensing costs associated with proprietary AI solutions. Furthermore, the system operates efficiently on standard personal computers without requiring high-end GPU infrastructure. This ensures cost-effective deployment in educational institutions, physiotherapy clinics, and home environments. The low implementation cost increases scalability and adoption potential.

**Environmental Constraints:**

As a software-centric solution, the project does not demand specialized hardware components, thereby reducing material consumption and electronic waste generation. The lightweight architecture minimizes energy consumption during runtime. The system's optimized processing pipeline reduces computational redundancy, contributing to lower power usage and environmentally responsible operation.

**Social Constraints:**

The system contributes positively to society by promoting health awareness and preventive care. By enabling real-time posture monitoring, the framework supports individuals in maintaining proper body mechanics during exercise and daily activities. The accessibility of the system encourages home-based rehabilitation, which benefits elderly individuals and patients recovering from injury. Additionally, the system fosters digital inclusion by requiring minimal hardware investment.

**Ethical Constraints:**

The project aligns with ethical AI standards by prioritizing user privacy and data security. Only skeletal landmark coordinates are processed; facial recognition or biometric identity data is neither stored nor transmitted. The system avoids bias by focusing on geometric analysis rather than demographic attributes. This ensures fair and responsible deployment.

Applicable standards:

- IEEE 7000 – Ethical Considerations in System Design
- IEEE 2857 – Privacy Engineering in Biometric Systems

**Health and Safety Constraints:**

The framework enhances user safety by detecting incorrect postures that may lead to long-term musculoskeletal injuries. The joint angle validation module identifies excessive strain or improper alignment, thereby reducing the risk of injury during workouts. This makes the system valuable for physiotherapy monitoring and injury prevention.

**Manufacturability Constraints**

Although primarily software-based, the system is designed for easy integration into portable devices, embedded systems, or mobile applications. The modular architecture supports scalability and cross-platform deployment. Minimal hardware requirements ensure manufacturability without complex production constraints.

**Sustainability Constraints**

The proposed system supports sustainable healthcare innovation by promoting preventive posture correction. The computational efficiency ensures lower energy consumption during continuous operation. Furthermore, the digital nature of the system reduces dependence on physical therapy tools, contributing to long-term sustainability in digital health solutions

# ABSTRACT

Oral cancer remains one of the most significant global health challenges, with a high rate of late-stage diagnosis leading to severe complications and reduced survival rates. Early detection plays a crucial role in improving patient outcomes; however, traditional diagnostic methods often require specialized expertise and time-consuming procedures. To address this limitation, this project proposes a deep learning-based artificial intelligence system for the early detection of oral cancer using clinical images.

The proposed system leverages advanced convolutional neural network architectures such as ConvNeXt and EfficientNetV2, which are trained and fine-tuned on datasets of oral lesions. These models are capable of accurately distinguishing between malignant and normal conditions, achieving high classification performance. The system enhances reliability by incorporating Explainable AI techniques, specifically Grad-CAM, which visually highlights the most relevant regions in the input images. This improves transparency and helps medical professionals better understand the decision-making process of the AI model.

The entire solution is implemented as a user-friendly web application using Flask. Healthcare practitioners can upload clinical images, receive real-time predictions, and analyze probability scores generated by the model. The system also automatically generates comprehensive PDF reports containing patient details, diagnostic results, Grad-CAM heatmaps, and risk assessments. Additionally, secure email functionality is integrated to facilitate easy sharing of reports.

By combining deep learning, computer vision, and automated reporting into a single platform, the proposed system provides a scalable and efficient tool for assisting pathologists and healthcare professionals. It significantly reduces diagnostic time, enhances accuracy, and supports early intervention. Overall, this approach demonstrates the potential of AI-driven solutions in improving healthcare delivery and enabling better clinical decision-making in oral cancer diagnosis.


***Keywords—Oral Cancer Detection, Deep Learning, Machine Learning, Computer Vision, Digital Image Processing, ConvNeXt, EfficientNetV2, PyTorch, OpenCV, Grad-CAM, Explainable AI (XAI), Image Classification, Medical Image Analysis, Flask, Web Application, Real-Time Inference, PDF Report Generation, Email Integration***.

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# LIST OF ACADEMIC REFERENCE COURSES

| S. NO. | COURSE CODE | COURSE NAME |
|---|---|---|
| **1** | 211CSE1402 | Python Programming |
| **2** | 212CSE2304 | Machine Learning |
| **3** | 212CSE2303 | Software Engineering |
| **4** | 213CSE3301 | Deep Learning |
| **5** | 213CSE1301 | AI And ML |

# CHAPTER – I

# INTRODUCTION

## 1.1 Background and Motivation

Human motion analysis has become a critical component of modern intelligent systems, particularly in healthcare monitoring, sports performance evaluation, rehabilitation engineering, and interactive computing. Traditional movement assessment methods rely heavily on manual observation or wearable sensor systems, which may be expensive, intrusive, or inconvenient for users. With the rapid advancement of artificial intelligence and computer vision, real-time human pose estimation has emerged as a non-invasive and scalable alternative for analyzing body posture and joint movement.

Recent developments in deep learning have significantly improved the ability to detect anatomical landmarks from images and video sequences. Frameworks such as MediaPipe provide lightweight yet highly accurate models capable of identifying 33 body landmarks in real time. However, while landmark detection accuracy has improved, many systems still face challenges related to latency, geometric stability, and biomechanical validation. Most implementations focus solely on key point localization without incorporating structured joint angle analysis, which limits their applicability in fitness correction and rehabilitation monitoring.

The motivation behind this project is to design a real-time pose estimation system that not only detects landmarks efficiently but also performs reliable joint angle computation using mathematical modeling. By combining MediaPipe's optimized pose detection model with OpenCV's real-time processing capabilities and vector-based geometric analysis, the proposed framework aims to bridge the gap between theoretical pose estimation research and practical, deployable movement analysis systems.

## 1.2 Problem Statement

Oral cancer remains a major global health concern due to its high mortality rate when diagnosed at advanced stages. Early detection is essential for improving patient survival, yet traditional diagnostic methods such as biopsy and clinical examination are time-consuming, costly, and require expert intervention. In many regions, especially rural and resource-limited areas, access to early screening facilities is limited, leading to delayed diagnosis and poor outcomes.

Although deep learning techniques have shown promising results in medical image analysis, existing systems often face practical limitations. High-accuracy models typically require significant computational resources, reducing their real-time applicability. On the other hand, lightweight models may compromise accuracy and reliability. Furthermore, most current approaches lack interpretability, providing predictions without clear explanations, which reduces trust among healthcare professionals and limits clinical adoption.

Therefore, there is a need for an efficient and reliable AI-based system that can assist in the early detection of oral cancer using clinical images. The system should ensure high accuracy, low latency, and consistent performance under varying conditions. It must provide explainable results through visualization techniques and support automated report generation with secure sharing. Such a solution can enhance diagnostic efficiency, reduce workload, and improve overall healthcare outcomes.

## 1.3 Objectives of the Project

The primary objective of this project is to develop an intelligent and efficient system for early detection of oral cancer using clinical images. The system aims to leverage advanced deep learning techniques to accurately classify oral lesions into malignant and normal categories. It focuses on improving diagnostic accuracy while ensuring fast and reliable performance suitable for real-time applications.

Another key objective is to implement and utilize state-of-the-art deep learning models such as EfficientNetV2 and ConvNeXt for robust feature extraction and classification. The project also aims to optimize model performance to work efficiently on standard hardware with minimal computational latency.

The project seeks to enhance transparency and trust in AI predictions by integrating Explainable AI techniques such as Grad-CAM. This allows the system to highlight important regions in the input images, enabling medical professionals to understand the reasoning behind the model's decisions.

Another important objective is to design and develop a user-friendly web application using Flask that allows users to upload images, receive real-time predictions, and view analysis results. The system also aims to automate the generation of detailed PDF reports that include patient information, prediction results, probability scores, and visual explanations.

Additionally, the project focuses on integrating secure email functionality to enable easy sharing of diagnostic reports. It aims to ensure data privacy and secure handling of patient information throughout the system.

The overall objective is to create a scalable, cost-effective, and practical solution that assists healthcare professionals in early diagnosis, reduces manual workload, and improves clinical decision-making. The system is intended to bridge the gap between advanced AI research and real-world medical applications, ultimately contributing to better healthcare outcomes.

## 1.4 Scope of the Project

The scope of this project focuses on the development of an AI-based system for the early detection of oral cancer using clinical images. The system is designed to assist healthcare professionals by providing automated analysis and classification of oral lesions into malignant and normal categories. It primarily targets improving early diagnosis, which is crucial for effective treatment and increased survival rates.

The project includes the implementation of advanced deep learning models such as EfficientNetV2 and ConvNeXt for accurate feature extraction and image classification. It also incorporates Explainable AI techniques like Grad-CAM to provide visual interpretation of predictions, making the system more transparent and trustworthy for medical use.

The system is developed as a web-based application using Flask, allowing users to upload clinical images and receive real-time diagnostic results. It supports automated PDF report generation that includes patient details, prediction outcomes, probability scores, and visual heatmaps. Additionally, the system enables secure sharing of reports through email integration.

The scope also covers image preprocessing, model inference, and visualization using computer vision techniques. The system is designed to work efficiently on standard computing devices, ensuring accessibility and scalability. It can be extended to support deployment in hospitals, diagnostic centers, and remote healthcare environments.

However, the system is intended to act as a supportive diagnostic tool and not a replacement for professional medical judgment. It relies on the quality of input images and trained datasets, which may affect performance in certain cases. Future enhancements may include multi-class classification, integration with electronic health records, and deployment on mobile or cloud platforms.

Overall, the project aims to provide a practical, efficient, and scalable solution that bridges the gap between AI technology and real-world healthcare applications, contributing to improved diagnostic processes and patient care.

**1.5 Methodology Overview:**

The proposed system follows a structured methodology to develop an efficient and reliable AI-based solution for oral cancer detection using clinical images. The methodology begins with data acquisition, where clinical images of oral lesions are collected from relevant datasets. These images include both malignant and normal cases, ensuring diversity in terms of lighting conditions, lesion types, and image quality.The next step involves data preprocessing, where images are standardized to a fixed resolution (224×224 pixels) to match the input requirements of deep learning models. Image normalization techniques are applied to scale pixel values based on ImageNet standards. Additional preprocessing steps such as format conversion and noise handling are performed to improve model performance.

Following preprocessing, the system utilizes advanced deep learning architectures such as EfficientNetV2 and ConvNeXt. These pretrained models are fine-tuned using transfer learning to adapt them to the oral cancer dataset. The models extract high-level features from input images and perform classification into cancerous and normal categories.Model training and evaluation are carried out using the PyTorch library. The dataset is divided into training and testing sets to evaluate performance metrics such as accuracy and prediction confidence. The system uses Softmax activation to generate probability scores for each class.Once trained, the model is integrated into a web-based application developed using Flask. The application provides an interface for users to upload clinical images and input patient details. Upon submission, the system processes the image and performs real-time inference to generate predictions.

To enhance interpretability, the methodology incorporates Explainable AI using Grad-CAM. This technique identifies and highlights the most relevant regions in the image that influence the model’s decision. The generated heatmaps are overlaid on the original image to provide visual explanations.The system also includes a reporting module that automatically generates detailed PDF reports. These reports contain patient information, prediction results, probability scores, Grad-CAM visualizations, and risk assessment summaries. This helps streamline the diagnostic workflow for healthcare professionals.

Additionally, an email integration module is implemented using secure SMTP protocols to send generated reports directly to users. This ensures efficient communication and easy sharing of diagnostic results.The methodology ensures efficient handling of image data, accurate model predictions, and seamless integration of AI with a user-friendly interface. The system is designed to operate on standard hardware with optional GPU acceleration for improved performance.

Finally, the entire workflow is tested under different conditions to ensure robustness, reliability, and scalability. The methodology bridges the gap between deep learning research and real-world healthcare applications by combining automation, interpretability, and accessibility in a single system.

### 1.6 Organization of the Report:

This report is organized into eight major chapters, each addressing a specific phase of the project development lifecycle, from conceptual foundation to implementation, evaluation, and conclusion.

**Chapter I – Introduction** provides the foundational context of the study. It presents the background and motivation behind oral cancer detection using artificial intelligence, outlines the problem statement, defines the objectives and scope of the project, and gives an overview of the methodology adopted. This chapter highlights the importance of early diagnosis and the role of AI in improving healthcare outcomes.

**Chapter II – Literature Review** examines existing research and related work in the field of medical image analysis and oral cancer detection. It reviews traditional diagnostic methods, deep learning-based classification techniques, and recent advancements in AI-driven healthcare systems. The chapter concludes with identified research gaps, particularly in interpretability and real-time deployment.

**Chapter III – System Analysis** presents the analytical foundation of the project. It describes functional and non-functional requirements of the system, including image processing, prediction accuracy, and user interaction. This chapter also includes feasibility analysis covering technical, operational, and economic aspects, along with risk analysis to identify challenges such as data variability and model reliability.

**Chapter IV – System Design** d details the architectural and structural design of the proposed system. It explains the overall system architecture, including image input, preprocessing, deep learning model inference, Grad-CAM visualization, and report generation modules. This chapter also presents workflow diagrams, system flow, and user interface design of the web application.

**Chapter V – Implementation** describes the technical execution of the project. It outlines the technology stack, including PyTorch, OpenCV, and Flask. The implementation of each module such as model loading, image processing, prediction, Grad-CAM generation, PDF report creation, and email integration is explained in detail.

**Chapter VI – Testing** presents the testing strategies used to validate system performance. It includes unit testing, integration testing, and system testing. Test cases are designed to verify image upload, prediction accuracy, report generation, and email functionality, ensuring system reliability and robustness.

**Chapter VII – Results and Discussion** analyses the outputs generated by the system. It includes evaluation metrics such as accuracy, confidence scores, and processing time. The performance of models like ConvNeXt and EfficientNetV2 is discussed, along with challenges such as dataset limitations and improvements made during development.

**Chapter VIII – Conclusion and Future Scope** summarizes the achievements of the project and highlights its contributions to AI-based healthcare solutions. It also discusses future enhancements such as multi-class classification, mobile or cloud deployment, integration with hospital systems, and improved dataset expansion for better accuracy.

Finally, the report concludes with the **References** section listing all academic sources consulted, followed by documentation of **Publications**, **Certifications**, and the **Plagiarism Report** as required by institutional guidelines.

# CHAPTER-II
# LITERATURE REVIEW

## 2.1 Overview of Related Work

Oral cancer detection has gained significant research attention due to its increasing prevalence and the critical importance of early diagnosis. Traditional diagnostic approaches rely on clinical examination, biopsy, and histopathological analysis, which are time-consuming and require expert intervention. With the advancement of artificial intelligence, particularly deep learning, automated medical image analysis has emerged as a promising solution for early-stage detection.

Initial research in this domain focused on applying convolutional neural networks (CNNs) for image classification tasks. These models demonstrated the capability to extract meaningful features from medical images and classify them into normal and abnormal categories. Over time, more advanced architectures such as EfficientNetV2 and ConvNeXt have been introduced, significantly improving classification accuracy and computational efficiency.

Recent studies have also emphasized the importance of Explainable AI in medical applications. Techniques such as Grad-CAM are widely used to visualize important regions in medical images, enabling better understanding of model predictions. These visualization methods help bridge the gap between AI systems and clinical trust by providing interpretable results.

Furthermore, the development of web-based AI systems has enabled real-time deployment of diagnostic tools. Frameworks such as Flask allow integration of deep learning models into user-friendly applications where healthcare professionals can upload images and receive instant predictions. These advancements highlight the shift toward accessible, scalable, and intelligent healthcare solutions.

## 2.2 Review of Similar Projects or Research Papers:

A systematic review on deep learning applications in oral cancer by Kritsasith Warin and Siriwan Suebnukarn highlights the effectiveness of AI models in improving early diagnosis and clinical decision-making. The study emphasizes the growing importance of deep learning in medical imaging while identifying challenges related to dataset quality and model generalization. [1] A tailored deep learning approach using a 19-layer CNN for oral cancer detection demonstrated high classification accuracy on clinical lip and tongue images. The model showed strong performance in early-stage detection, though it required optimized preprocessing for better robustness. [2] An AI-based oral cancer detection system integrated with a portable electronic oral endoscope enabled real-time image acquisition and diagnosis. This approach improved accessibility in remote healthcare settings while maintaining reliable detection accuracy. [3] An optimized deep learning ensemble model combined multiple architectures to improve classification accuracy. The study showed that ensemble techniques significantly enhance performance compared to single-model approaches. [4]

A hybrid model using Deep Belief Networks and group teaching optimization achieved improved diagnostic accuracy. This method demonstrated the effectiveness of combining optimization algorithms with deep learning for medical applications. [5] A comprehensive review of AI techniques for oral cancer detection analyzed multiple machine learning and deep learning approaches. It highlighted the advantages of automated systems while addressing issues such as data imbalance and lack of standard evaluation metrics. [6]

AI-based systems for oral cancer diagnosis and risk assessment have shown strong potential in improving early detection rates. These systems provide decision support tools for clinicians, though interpretability remains a key concern. [7]Transfer learning approaches using microscopic images have proven effective in oral cancer detection. Pretrained models significantly reduce training time while maintaining high accuracy. [8]A deep learning model integrated with LIME explainable AI technique provided improved interpretability in oral cancer classification. This approach helped visualize model decisions, enhancing trust among medical professionals. [9]The proposed OCU-Net architecture, based on U-Net, improved segmentation accuracy of oral cancer regions. This model effectively captured spatial features and enhanced lesion boundary detection. [10]Explainable AI-based multiclass classification using Grad-CAM visualization demonstrated improved interpretability in histopathology image analysis. The study showed how visual explanations support clinical validation. [11]Smartphone-based and cloud-based AI tools have enabled real-time screening of oral cancer and potentially malignant disorders. These systems improve accessibility and scalability in low-resource environments. [12]Advanced deep learning algorithms have been widely applied in oral cancer detection, showing improvements in accuracy, efficiency, and automation. These techniques highlight the growing role of AI in healthcare. [13]

Automated classification systems for oral cancer diagnosis have demonstrated high accuracy using clinical images. However, challenges remain in handling diverse datasets and varying image conditions. [14]Deep learning-based classification combined with Grad-CAM visualization improved interpretability and diagnostic performance. This approach provides both prediction and explanation in a unified framework. [15] CapsNet and Deep Belief Network-based approaches have been used for detecting oral leukoplakia. These models capture spatial hierarchies and improve feature representation for better classification. [16] Diffusion models have been applied to generate synthetic lesions, improving dataset diversity and diagnostic accuracy. This technique helps overcome data scarcity in medical imaging. [17]

Studies on diffusion models for skin and oral cancer classification demonstrate their ability to enhance model robustness and generalization. [18] AI-based systematic reviews on oral squamous cell carcinoma highlight the role of machine learning in improving diagnosis and clinical workflows. These studies emphasize the need for scalable and interpretable systems. [19] Machine learning combined with dimensionality reduction techniques has shown improved efficiency and accuracy in oral cancer detection systems. [20] Multi-modal approaches using weighted ensemble CNNs integrate multiple data sources to enhance detection performance. These systems provide better generalization and robustness. [21]Optimized ensemble learning models further improve classification accuracy by combining predictions from multiple networks, reducing model bias and variance. [22]CapsNet-based hybrid models continue to demonstrate improved feature extraction and classification performance in oral cancer detection tasks. [23]Diffusion-based approaches for cancer classification enhance data augmentation and improve deep learning model performance. [24]Multi-modal ensemble CNN models provide improved accuracy by leveraging complementary features from different input modalities. [25]Ensemble-based deep learning approaches consistently achieve higher performance compared to individual models in oral cancer classification tasks. [26]Advanced hybrid architectures combining CapsNet and deep learning techniques improve classification reliability and feature learning. [27]Diffusion models continue to show promise in enhancing classification performance and dataset diversity in oral cancer research. [28]

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### 2.3 Summary and Gap Identification:

The reviewed literature demonstrates substantial progress in real-time pose estimation and exercise posture correction using deep learning frameworks such as MediaPipe and BlazePose. Many systems successfully achieve high detection accuracy and real-time responsiveness. Furthermore, integration with machine learning models enables classification of exercises and detection of incorrect movements.

However, several research gaps remain evident. First, many existing systems prioritize repetition counting or activity classification rather than detailed biomechanical joint angle validation. Although joint angles are sometimes computed, few studies emphasize geometric stability analysis using mathematical modeling principles. Second, while lighting enhancement and adaptive preprocessing techniques are explored in some works, systematic evaluation of latency reduction alongside accuracy improvement is rarely addressed.

Third, most systems lack comprehensive benchmarking using multiple datasets, especially custom exercise-specific datasets designed to test robustness under varied environmental conditions. Additionally, limited attention has been given to statistical validation of improvements over baseline implementations. Finally, concerns regarding privacy, scalability, and deployment on standard hardware remain insufficiently explored in many prior works.

The proposed system addresses these gaps by integrating real-time landmark detection with vector-based joint angle computation, latency optimization, and multi-dataset benchmarking. Unlike previous works that focus primarily on repetition recognition, this framework emphasizes geometric modeling, biomechanical validation, and statistical evaluation. By combining accuracy enhancement with computational efficiency, the proposed system aims to provide a more comprehensive and scalable solution for posture monitoring and intelligent motion analysis.

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# CHAPTER III
# SYSTEM ANALYSIS

## 3.1 Requirements Gathering

The requirements for the proposed oral cancer detection system were identified through systematic evaluation of user needs, technical constraints, and healthcare application objectives. The primary stakeholders considered during the requirement analysis include healthcare professionals, pathologists, medical researchers, and patients seeking early diagnostic support. The system is designed to assist medical practitioners in analyzing clinical images for early detection of oral cancer, thereby improving diagnostic accuracy and reducing decision-making time.

The requirements gathering process involved analyzing existing AI-based medical imaging systems, reviewing literature on oral cancer detection using deep learning, and identifying limitations in current solutions. Special emphasis was placed on achieving high classification accuracy, real-time inference, and reliable performance under varying image conditions. Advanced deep learning models such as EfficientNetV2 and ConvNeXt were considered to ensure robust feature extraction and accurate prediction.

Additionally, the system is designed to function as a web-based application using Flask, enabling users to upload images and receive instant diagnostic results. The system also incorporates Explainable AI techniques such as Grad-CAM to provide visual interpretation of predictions, improving transparency and user trust.

Key considerations during requirements analysis include prediction accuracy, processing speed, interpretability of results, user interface usability, data privacy, and system scalability. The system must also support automated PDF report generation and secure email sharing to streamline clinical workflows. Furthermore, it is designed to operate efficiently on standard computing devices without requiring specialized hardware, ensuring accessibility and cost-effectiveness.

These requirements collectively guided the design and development of the proposed system, ensuring a balance between performance, usability, and practical deployment in real-world healthcare environments.

## 3.2 Functional Requirements

The functional requirements define the specific operations that the system must perform to achieve its intended objectives. The system must be capable of capturing real-time video input through a connected camera device and processing individual frames continuously without noticeable delay. It must accurately detect 33 anatomical landmarks using MediaPipe's pose estimation framework and extract joint coordinate data from each frame.The system must compute joint angles using vector-based mathematical modeling and compare calculated values against predefined biomechanical thresholds to determine posture correctness. When incorrect alignment is detected, the system must generate real-time visual feedback by overlaying angle values and posture status indicators on the video stream.Additionally, the system must maintain performance evaluation capability by recording metrics such as accuracy, mean squared error, and processing latency. It should allow users to perform multiple exercises including squats, push-ups, standing, sitting, and walking while ensuring stable landmark tracking under varying lighting conditions.

The framework must also support dataset benchmarking using standard datasets such as COCO Keypoints and MP11, as well as the custom Exercise Pose Dataset (EPD 2026), enabling validation of system robustness.

### 3.3 Non-Functional Requirements

Non-functional requirements define the quality attributes that the system must satisfy. One of the most critical non-functional requirements is real-time performance. The system must maintain frame processing latency below acceptable thresholds (preferably under 40 milliseconds per frame) to ensure smooth visualization at approximately 25–35 frames per second.

Accuracy is another essential non-functional requirement. The system should achieve posture classification accuracy above 90% and maintain low mean squared error in joint angle computation. Stability under dynamic motion conditions and robustness to moderate lighting variations are also important quality attributes.

Scalability is required to allow potential expansion toward multi-person detection or 3D pose analysis in the future. The system must be modular in design, enabling independent modification of components such as detection engine, angle computation module, and visualization interface.

Security and privacy must also be maintained. The system should process only skeletal landmark data without storing personally identifiable information. Finally, usability and user interface clarity are essential to ensure that posture feedback is understandable and actionable for end users.

### 3.4 Feasibility Study

A feasibility study was conducted to evaluate whether the proposed oral cancer detection system can be successfully implemented within technical, operational, and economic constraints. The analysis confirms that the system is feasible due to the availability of open-source technologies, pretrained deep learning models, and standard computing resources.

From a **technical perspective**, the system is feasible as it utilizes widely adopted frameworks such as PyTorch, OpenCV, and Flask. These tools support efficient model development, image processing, and web deployment. The use of pretrained architectures like EfficientNetV2 and ConvNeXt further simplifies implementation and ensures high performance without requiring extensive computational resources.

From an **operational perspective**, the system is user-friendly and can be easily adopted by healthcare professionals. It provides a simple interface for uploading images, generating predictions, and obtaining reports. The integration of automated PDF generation and email sharing enhances usability and streamlines clinical workflows.

From an **economic perspective**, the system is cost-effective as it relies entirely on open-source software and does not require specialized hardware. It can run on standard computers, making it accessible to hospitals, clinics, and research institutions with minimal investment.

Overall, the feasibility study indicates that the proposed system is practical, scalable, and suitable for real-world deployment in healthcare environments.

### 3.4.1 Technical Feasibility

The technical feasibility of the system is strong due to the availability of mature open-source libraries such as MediaPipe and OpenCV. MediaPipe provides optimized landmark detection capable of real-time performance on standard hardware, while OpenCV offers efficient video capture and image processing capabilities.

The computational requirements are moderate, and the system can operate on personal computers equipped with standard processors and at least 8GB of RAM. Optional GPU acceleration can further enhance performance but is not mandatory. The mathematical modeling required for joint angle computation is computationally lightweight, making implementation technically practical.

Furthermore, the modular architecture allows independent testing and refinement of each subsystem, reducing implementation complexity. Therefore, from a technical standpoint, the project is feasible and implementable using currently available technologies.

### 3.4.2 Operational Feasibility

Operational feasibility evaluates whether the system can function effectively in real-world healthcare environments and whether users can easily adopt it. The proposed system is designed as a web-based application that requires only a standard computer or laptop with internet access, making it suitable for use in hospitals, clinics, diagnostic centers, and even remote healthcare settings.

The user interface is developed to be simple, intuitive, and user-friendly, allowing healthcare professionals to upload clinical images, view predictions, and generate reports with minimal effort. The system does not require specialized hardware or complex setup, which reduces operational complexity and enhances usability.

Additionally, the system integrates multiple functionalities such as image analysis, prediction, Grad-CAM visualization, PDF report generation, and secure email sharing within a single platform. This streamlines the diagnostic workflow and reduces manual effort for healthcare practitioners.

The system is flexible and can handle images with varying quality, lighting conditions, and lesion characteristics, ensuring reliable performance across different clinical scenarios. It can also be easily adapted for future enhancements such as mobile or cloud deployment.

Therefore, the proposed system demonstrates strong operational feasibility by providing an accessible, efficient, and practical solution for real-world oral cancer detection and clinical support.

### 3.4.3 Economic Feasibility

The proposed system is economically viable due to its reliance on open-source software frameworks and minimal hardware requirements. Technologies such as PyTorch, OpenCV, and Flask are freely available, eliminating the need for costly software licenses.

The system requires only a standard computing device for deployment, which is commonly available in hospitals, clinics, and research environments. No specialized medical imaging hardware or additional equipment is required, reducing initial investment and operational costs.

Development costs are primarily limited to software design, model training, and system testing. Since the system is fully software-based, there are no manufacturing or hardware maintenance expenses involved. Additionally, the modular design allows easy upgrades and scalability without significant financial overhead.

As a result, the proposed system is cost-effective and can be widely deployed in healthcare institutions, diagnostic centers, and remote medical facilities. Its affordability and scalability make it a practical solution for improving early oral cancer detection and supporting digital healthcare initiatives.

## 3.5 Risk Analysis

Although the proposed system is feasible, certain risks must be considered during implementation. One potential risk is reduced prediction accuracy due to poor image quality, such as low lighting, blur, or low resolution. This can affect the performance of deep learning models and lead to incorrect classification. To mitigate this risk, proper image preprocessing techniques and user guidelines for capturing clear clinical images can be implemented.

Another risk involves increased processing time when handling high-resolution images or large datasets. This may impact real-time performance and user experience. This issue can be addressed through image resizing, model optimization, and efficient memory management techniques to ensure faster inference.

Variability in dataset quality and diversity is another important concern. If the training data does not include sufficient variations in lesion types, lighting conditions, and patient demographics, the model’s generalization ability may be limited. This risk can be mitigated by expanding the dataset and incorporating diverse samples during training.

Privacy and data security also represent potential risks. Since the system processes clinical images and patient information, it is essential to ensure secure handling of data. This can be addressed by avoiding unnecessary data storage, using secure communication protocols, and implementing proper access control mechanisms.

Hardware limitations may affect system performance on low-end devices, particularly during model inference. This can be mitigated by using optimized models and allowing optional GPU acceleration when available.

Overall, the identified risks are manageable and can be effectively mitigated through careful system design, optimization strategies, and adherence to best practices in AI-based healthcare applications.

# CHAPTER IV

# SYSTEM DESIGN

## 4.1 Overall System Architecture:

The overall architecture of the proposed oral cancer detection system follows a layered and modular design approach to ensure scalability, maintainability, and performance efficiency. The architecture consists of five primary layers: Input Acquisition Layer, Preprocessing Layer, Model Inference Layer, Explainability Layer, and Output & Reporting Layer.

The **Input Acquisition Layer** is responsible for receiving clinical images uploaded by users through a web interface. The system validates file formats and securely stores the input image for further processing.

The **Preprocessing Layer** performs image transformations such as resizing, normalization, and format conversion to ensure compatibility with deep learning models. These steps standardize input images and improve model performance under varying conditions.

The **Model Inference Layer** utilizes advanced deep learning architectures such as EfficientNetV2 and ConvNeXt implemented using PyTorch. This layer extracts features from the input image and classifies it into cancerous or normal categories while generating probability scores.

The **Explainability Layer** integrates Grad-CAM (Gradient-weighted Class Activation Mapping) to identify and highlight the most relevant regions in the image that influence the model's prediction. This enhances transparency and helps medical professionals understand the reasoning behind the results.

The **Output & Reporting Layer** presents the final prediction, confidence score, and risk level to the user through a web interface developed using Flask. It also generates a detailed PDF report containing patient information, prediction results, and Grad-CAM visualizations. Additionally, the system supports secure email sharing of reports for efficient communication.

This modular architecture ensures efficient data flow, real-time performance, and easy scalability, making the system suitable for deployment in practical healthcare environments.

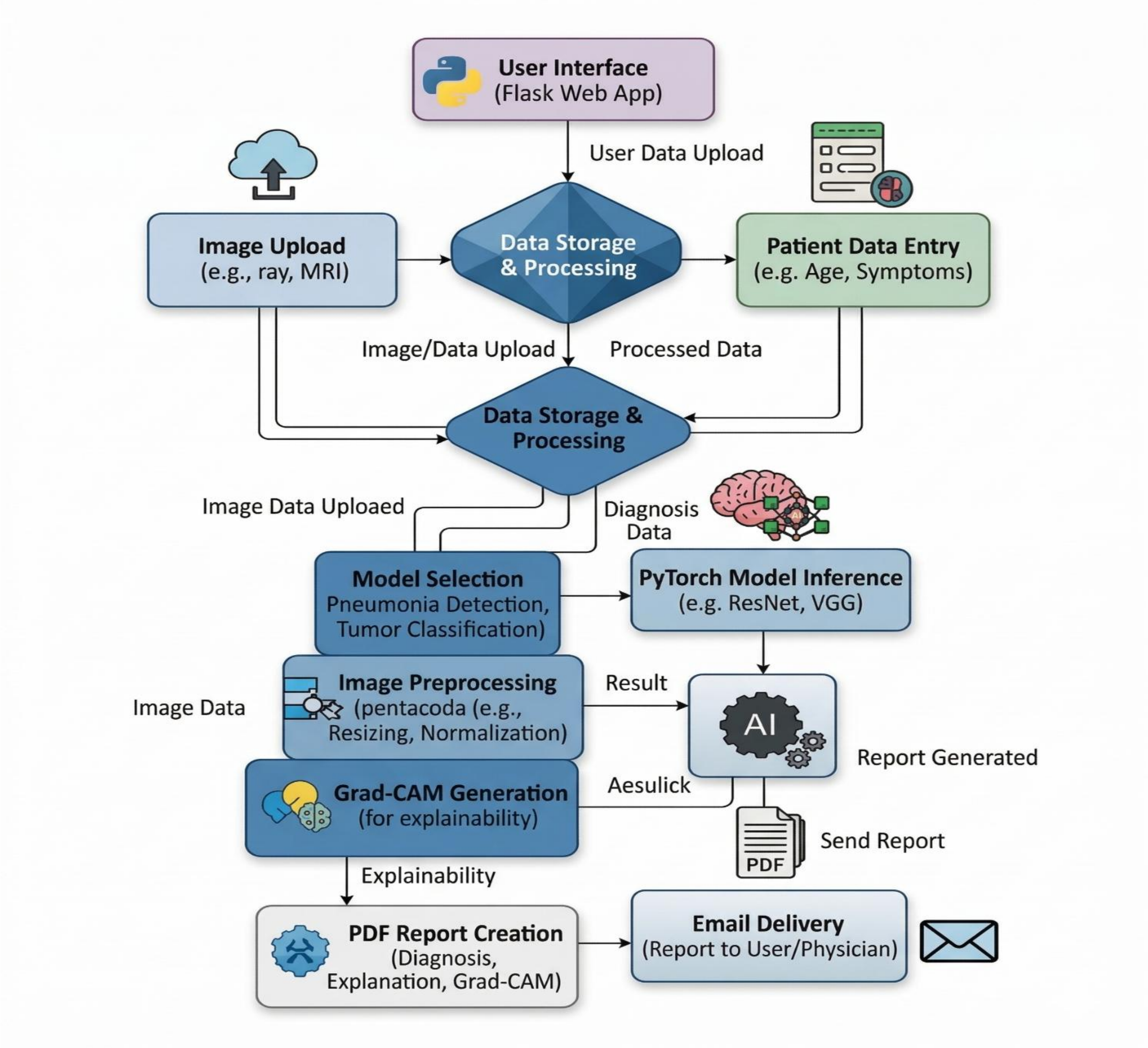


**Fig.1. Architecture of the system**

### 4.2 Module Design

The system is divided into independent modules to ensure modularity and simplified debugging. Each module performs a specific task while communicating with other modules through defined interfaces.

#### 4.2.1 Module 1 – Model Inference Module

The Model Inference Module is responsible for analyzing the input clinical image and performing classification. It utilizes deep learning models such as EfficientNetV2 and ConvNeXt. The module takes a preprocessed image as input and produces prediction results along with confidence scores, ensuring accurate and reliable detection.

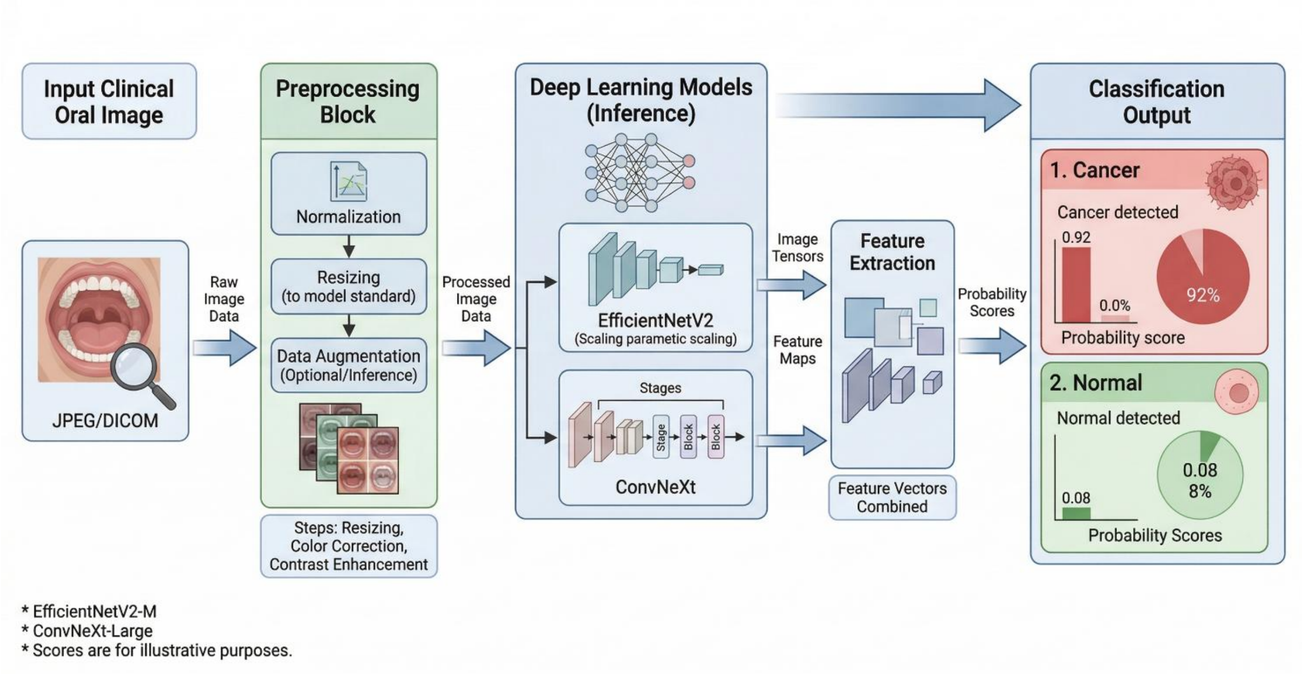


**Fig.2. Model Inference Module**

### 4.2.2 Module 2 – Image Preprocessing Module

This module processes the uploaded clinical image to prepare it for deep learning model input. It performs operations such as resizing the image to a standard resolution (224×224), normalization, and format conversion using libraries like NumPy and PIL.

The module ensures consistency in input data and improves model performance by reducing noise and variations caused by lighting and image quality. It also standardizes the image format, making it compatible with deep learning models such as EfficientNetV2 and ConvNeXt.

### 4.2.3 Module 3 – Explainability (Grad-CAM) Module

This module analyzes the model prediction and highlights important regions in the clinical image using Grad-CAM. It identifies the areas that contribute most to the classification result and generates a heatmap overlay.

This module transforms the system from a simple prediction tool into an interpretable AI system by providing visual explanations, thereby improving transparency and trust for medical professionals.

#### 4.2.4 Module 4 – Report Generation & Monitoring Module

This module records system outputs and generates structured reports based on the prediction results. It includes key metrics such as prediction confidence, classification accuracy, and processing time for performance evaluation.

The module generates a detailed PDF report containing patient details, prediction results, probability scores, and Grad-CAM visualizations. It also supports monitoring system performance and enables comparison with baseline models for evaluation.

### 4.3 Database Design :

Although the system primarily performs real-time image analysis, a lightweight data storage mechanism is used for managing user inputs, report files, and system outputs. The system stores uploaded images, generated PDF reports, and basic user details such as name and email for communication purposes.

The design avoids storing sensitive medical data permanently, ensuring privacy and security. File-based storage or minimal database integration is used to maintain simplicity, efficiency, and easy retrieval of generated reports when required.

#### 4.3.1 ER Diagram:

The Entity-Relationship (ER) diagram represents the logical structure of the system designed to store user inputs, diagnostic results, and report data for the oral cancer detection framework.

**User** – Represents an individual (patient or healthcare professional) who uses the system to upload clinical images and receive diagnostic reports.

**Image** – Represents the clinical image uploaded by the user for analysis.

**Analysis** – Stores the prediction results generated by the deep learning model, including classification (cancer/normal), confidence score, and risk level.

**Report** – Represents the generated PDF report containing patient details, prediction results, and Grad-CAM visualizations.

**Performance** – Stores system-related metrics such as processing time and prediction confidence for evaluation purposes.

**Relationships:**

- **User uploads Image** – A user can upload multiple clinical images.
- **Image generates Analysis** – Each image is processed to produce a prediction result.
- **Analysis generates Report** – Each analysis result produces a detailed PDF report.
- **Analysis records Performance metrics** – System performance data is associated with each prediction.

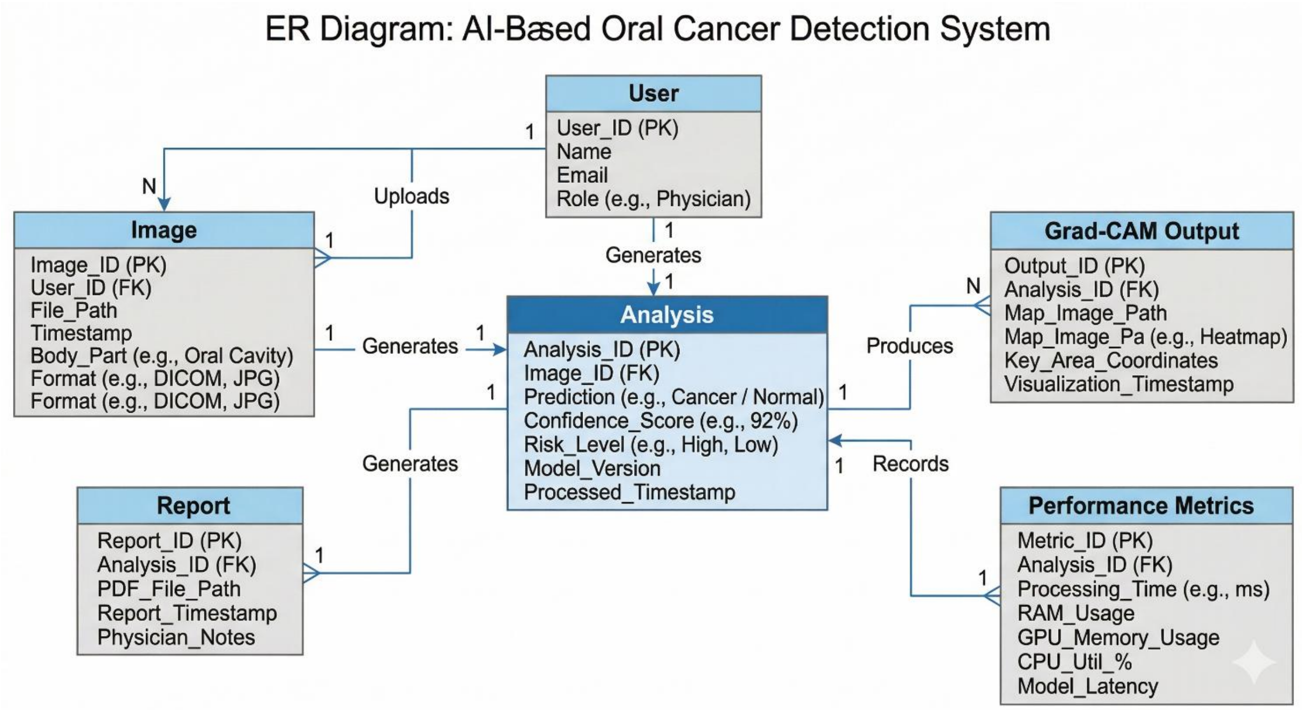


**Fig.3.Database Schema**

The database schema defines the physical structure of tables and their attributes required to store and manage system data efficiently.

**User** – Stores user details such as name, email, and basic identification information.

**Image Data** – Stores uploaded image file paths, timestamps, and associated user details.

**Analysis Result** – Stores prediction outputs including classification label, confidence score, and risk level.

**Grad-CAM Output** – Stores generated heatmap image paths for visualization.

**Report Data** – Stores generated PDF report file paths and related metadata.

**Performance Metrics** – Stores system metrics such as processing time and inference details for each analysis.

### 4.4 User Interface Design

The user interface of the proposed system is designed to ensure clarity, simplicity, and ease of interaction for healthcare professionals and users. The primary objective of the interface is to provide accurate diagnostic results in a clear and understandable manner without overwhelming the user with technical complexity. The main interface consists of an image upload section, patient detail input fields, and a results display panel.

Once an image is uploaded, the system processes it and displays the prediction results along with confidence scores. The interface also presents Grad-CAM visualizations, highlighting important regions in the clinical image to improve interpretability. A risk indicator is included to inform users about the severity level, making the output more meaningful for clinical use.The interface uses clean layouts and minimal design elements to enhance usability. Clear labels, structured output sections, and easy navigation ensure that users can interact with the system efficiently. The system also provides options to download the generated PDF report and send it via email.

The interface ensures fast response time and smooth interaction, maintaining system reliability and user engagement

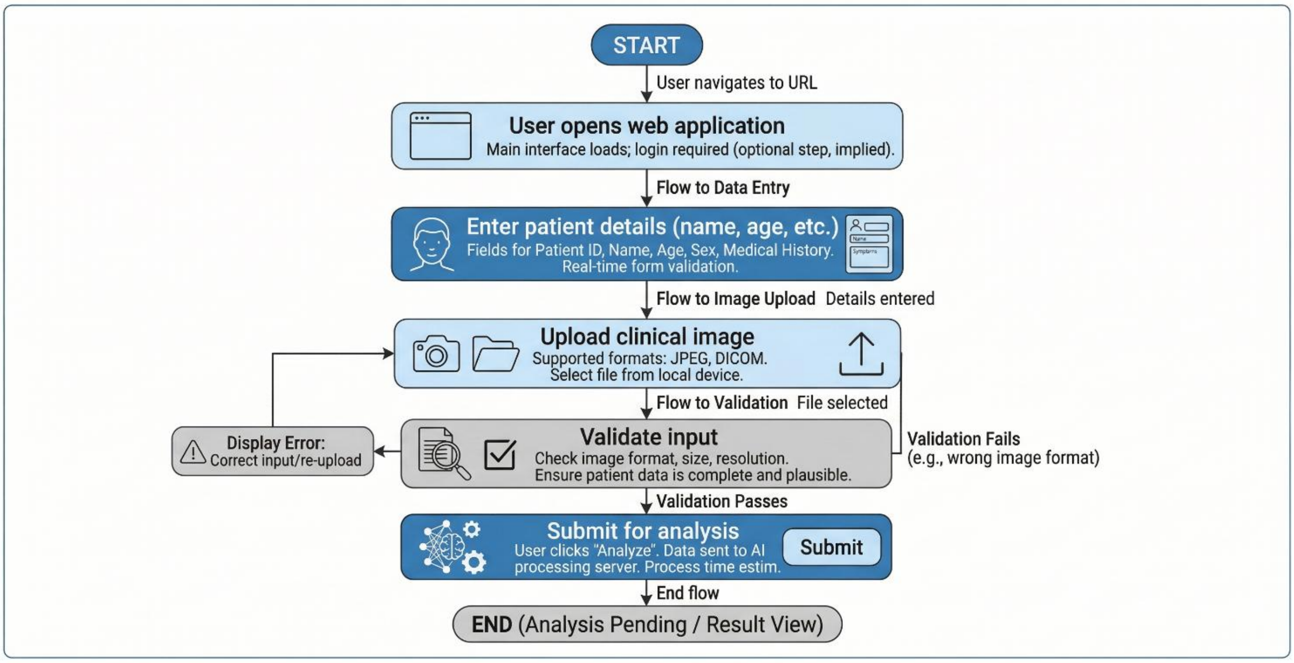


**Fig.4. User Flow Diagram 1 (Input Flow)**

This diagram illustrates the initial interaction flow where the user accesses the system, enters patient details, uploads a clinical image, and submits the data for analysis.

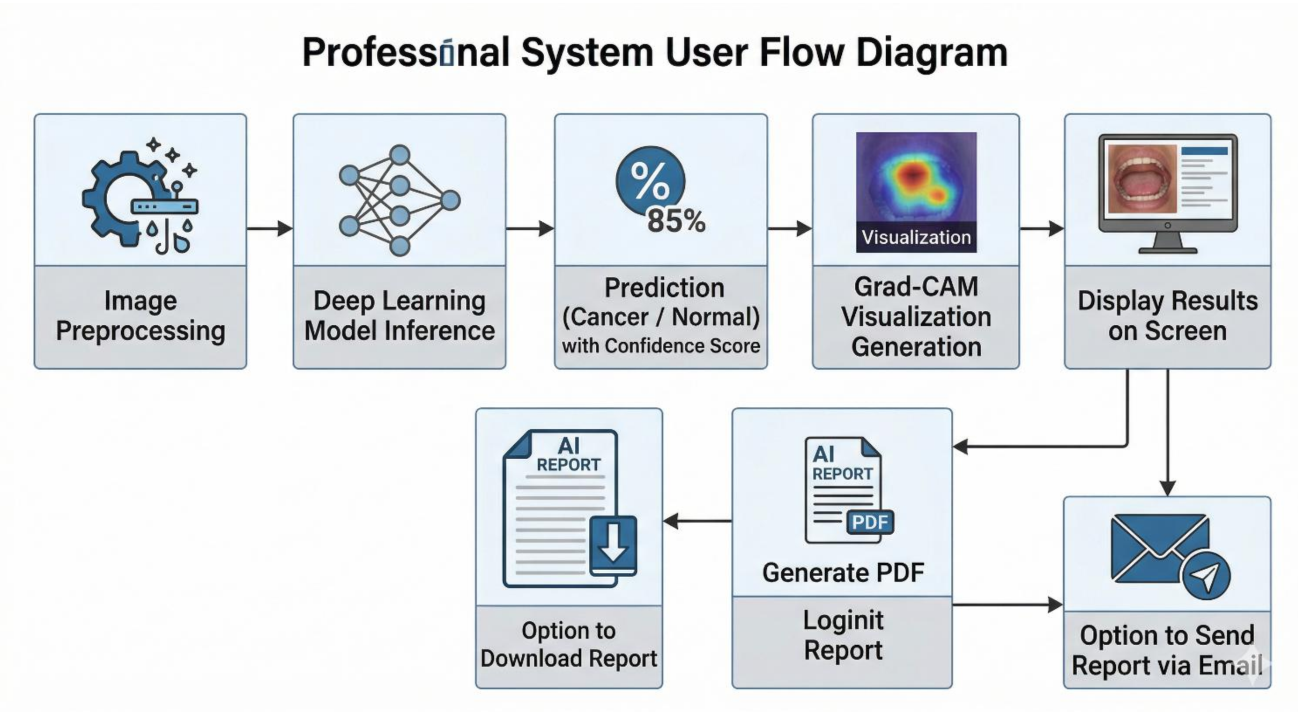


**Fig.5. User Flow Diagram 2 (Processing & Output Flow)**

This diagram represents the processing and output flow, including image preprocessing, model inference, Grad-CAM visualization, result display, PDF report generation, and optional email sharing of the report.

# CHAPTER V
# IMPLEMENTATION

## 5.1 Technology Stack:

The implementation of the proposed oral cancer detection system is carried out using a lightweight, scalable, and efficient software stack focused on real-time performance and ease of deployment. The system integrates deep learning frameworks, computer vision libraries, and web-based technologies to ensure smooth functionality and user interaction.

The core processing engine of the system is implemented in Python, which provides strong support for machine learning, image processing, and web application development. Python's modular structure allows seamless integration of various libraries and simplifies system development.

For deep learning-based image classification, frameworks such as PyTorch are used to implement and deploy models like EfficientNetV2 and ConvNeXt. These models enable accurate feature extraction and classification of clinical images.

For image processing and visualization tasks, OpenCV is utilized to handle image transformations and Grad-CAM heatmap generation. The system also uses libraries such as NumPy and PIL for efficient image manipulation and preprocessing.

The web interface is developed using Flask, which enables the system to function as a browser-based application. It allows users to upload images, view predictions, and generate reports without requiring complex frontend development.

Additional components such as PDF generation (FPDF) and email integration (SMTP) are included to automate report creation and communication. This technology stack ensures that the system is platform-independent, scalable, and capable of running efficiently on standard computing devices.

### 5.1.1 Programming Languages and Tools

The primary programming language used in the project is **Python**, chosen for its extensive ecosystem of libraries related to artificial intelligence and computer vision. Python facilitates mathematical modelling, real-time frame processing, and integration with web frameworks.

The key tools and libraries used in the implementation are:

- **Media Pipe** – For real-time human pose detection and landmark extraction.
- **OpenCV** – For video capture, frame manipulation, and visualization overlay.
- **NumPy** – For efficient numerical operations and vector-based joint angle calculations.
- **Stream lit** – For launching the interactive web application interface.
- **Matplotlib (optional)** – For visualizing performance metrics and evaluation graphs.

The development environment includes Visual Studio Code and Anaconda for dependency management. The system is designed to operate on Windows, Linux, or macOS platforms.

## 5.2 Implementation of Modules

The system is implemented in a modular fashion to ensure maintainability and scalability. Each module is independently developed and tested before being integrated into the final application.

### 5.2.1 Module 1 – Image Input Module

The Image Input Module is responsible for accepting clinical images uploaded by the user through the web interface. It validates the input file format (such as JPG, JPEG, or PNG) and securely stores the image for further processing within the system.

The implementation begins by receiving the uploaded image through the Flask-based web application. The file is checked for validity and saved using secure file handling mechanisms. The image is then loaded and converted into the appropriate format (RGB) using libraries such as PIL.

The module ensures that the input image is properly prepared for preprocessing and model inference. It handles different image sizes and formats while maintaining consistency in data flow. Efficient file handling and validation techniques are applied to ensure system reliability and prevent errors during execution.

This module is designed to operate efficiently with minimal delay, ensuring smooth interaction and seamless integration with subsequent modules in the system pipeline.

### 5.2.2 Module 2 – Image Preprocessing Module

The Image Preprocessing Module processes the uploaded clinical image to prepare it for deep learning model inference. It ensures that the input image is standardized and suitable for accurate classification.

The module performs operations such as resizing the image to a fixed resolution (224×224), normalization using predefined mean and standard deviation values, and conversion into tensor format. These steps are implemented using libraries such as NumPy and torchvision transforms.

The preprocessing ensures consistency across different input images and reduces variations caused by lighting conditions, resolution differences, and noise. This improves the performance and reliability of the deep learning models.

Additionally, the module ensures that the processed image is efficiently passed to the model inference module without unnecessary computational overhead, maintaining overall system efficiency and stability.

### 5.2.3 Module 3 – Model Inference Module

The Model Inference Module analyzes the preprocessed clinical image to generate classification results. It utilizes deep learning models such as EfficientNetV2 and ConvNeXt implemented using PyTorch.

The module receives the processed image tensor as input and performs forward propagation through the trained model to extract features and classify the image into cancerous or normal categories. The output is passed through a Softmax function to obtain probability scores for each class. Based on the predicted class and confidence score, the system determines the final result and associated risk level. The module ensures consistent and accurate predictions while handling variations in input images.

The classification results and confidence values are then forwarded to subsequent modules such as Grad-CAM visualization and report generation for further processing.

### 5.2.4 Module 4 – Web Interface Module

The Web Interface Module is responsible for launching the system as an interactive web application. The interface is developed using Flask, which enables smooth interaction between the user and the backend system.

The interface includes components such as HTML forms for entering patient details, file upload options for clinical images, and result display sections for showing predictions and visual outputs. It allows users to upload images, submit data, and view results including classification, confidence scores, and Grad-CAM visualizations.

The application runs locally in a web browser, providing a simple and user-friendly interface without requiring complex frontend technologies. The Flask framework ensures efficient routing, request handling, and integration with backend processing modules, delivering a seamless user experience.

### 5.2.5 Module 5 – Grad-CAM Module

The Grad-CAM Module is responsible for providing visual explanations of the model's predictions. It identifies the regions in the clinical image that contribute most to the classification decision. This is achieved using Gradient-weighted Class Activation Mapping (Grad-CAM), implemented with the help of PyTorch.

The module extracts feature maps from the final convolutional layer of the model and computes gradients with respect to the predicted class. These gradients are used to generate a heatmap that highlights important areas in the image. The heatmap is then resized and overlaid on the original image using OpenCV.

This module enhances interpretability and helps medical professionals understand the reasoning behind the model's decision, thereby improving trust and usability of the system.

### 5.2.6 Report Generation Module

The Report Generation Module is responsible for creating a structured and detailed PDF report based on the analysis results. It uses the FPDF library to generate professional reports that include patient details, prediction results, confidence scores, risk level, and Grad-CAM visualizations.

The module organizes information into clearly defined sections such as patient information, diagnostic results, and visual evidence. It embeds both the original clinical image and the Grad-CAM output into the report for better understanding.

The generated report is saved in the system and made available for download through the web interface. This module helps streamline documentation and supports clinical decision-making by providing well-structured diagnostic outputs.

### 5.2.7 Email Module

The Email Module is responsible for sending the generated PDF report to the user or healthcare professional. It utilizes SMTP protocols through Python's smtplib library to securely send emails.

The module attaches the generated PDF report and sends it to the specified email address provided by the user. It uses secure authentication and encrypted connections to ensure safe communication.

This functionality enables easy sharing of diagnostic results and enhances system usability by allowing remote access to reports. It also supports faster communication between patients and medical professionals.

## 5.3 Integration of Modules

The integration phase combines all independently developed modules into a unified system to ensure seamless operation and efficient data flow. Each module interacts through well-defined interfaces, enabling smooth communication and reliable performance.

The workflow begins when the user accesses the web application developed using Flask. The user enters patient details and uploads a clinical image through the interface. The Image Input Module receives and validates the uploaded file, which is then passed to the Image Preprocessing Module for resizing, normalization, and format conversion.

The preprocessed image is forwarded to the Model Inference Module, where deep learning models such as EfficientNetV2 and ConvNeXt generate classification results along with confidence scores. These results are then sent to the Grad-CAM Module, which produces visual explanations by highlighting important regions in the image.

The outputs from these modules are passed to the Report Generation Module, where a detailed PDF report is created. The report is then made available for download and optionally sent to the user via the Email Module.

This sequential data flow ensures efficient processing, real-time response, and smooth interaction between system components. The modular integration approach maintains loose coupling, enabling easy debugging, maintenance, and future scalability. The final integrated system delivers accurate predictions, interpretable results, and automated reporting in a unified platform.

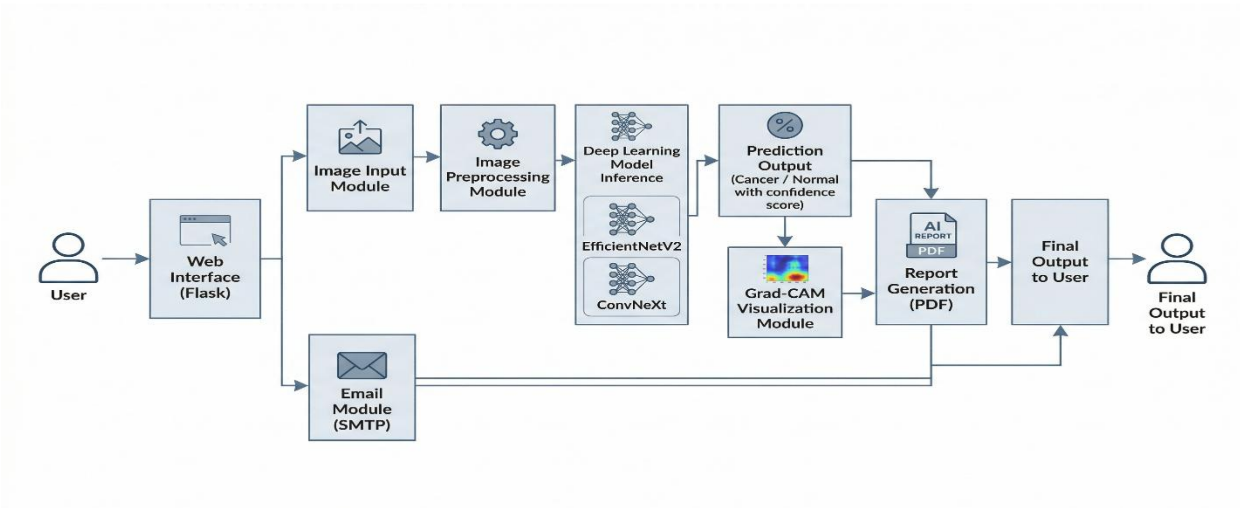


**Fig.6.1ntegration of Modules**

# CHAPTER VI

# TESTING

## 6.1 Testing Methodology

Testing is a critical phase in the development lifecycle to ensure that the oral cancer detection system performs accurately, efficiently, and reliably under different conditions. The testing strategy adopted in this project follows a multi-level validation approach including unit testing, integration testing, system testing, and user acceptance testing.

The testing process focused on validating image classification accuracy, model inference reliability, Grad-CAM visualization correctness, and overall system performance. Evaluation was conducted using clinical image datasets containing both cancerous and normal cases. Performance metrics such as Accuracy, Precision, Recall, F1-Score, and processing time were measured to validate system effectiveness.

The system was tested under varying conditions such as differences in image quality, lighting, resolution, and lesion characteristics to ensure robustness and reliability.

### 6.1.1 Unit Testing

Unit testing was conducted to verify the correctness of individual modules before integration.

The Image Input Module was tested to ensure proper file upload, validation, and storage of clinical images. The Image Preprocessing Module was validated by checking correct resizing, normalization, and tensor conversion.

The Model Inference Module was tested to ensure accurate classification and correct generation of probability scores. The Grad-CAM Module was verified by ensuring proper heatmap generation and correct overlay on input images.

Each module was tested independently to ensure logical correctness, stability, and error handling.

### 6.1.2 Integration Testing

Integration testing ensured that all modules functioned cohesively when combined.

The uploaded image from the Image Input Module was passed to the Preprocessing Module, and the processed image was forwarded to the Model Inference Module. The prediction results were then sent to the Grad-CAM Module for visualization, followed by the Report Generation Module for PDF creation.

The primary objective was to ensure smooth data flow and minimal processing delay across modules. Integration testing confirmed that the system operates without errors and maintains consistent performance.

### 6.1.3 System Testing

System testing evaluated the complete application under realistic operating conditions.

The system was tested using multiple clinical images with varying characteristics to assess performance. The evaluation measured classification accuracy, prediction confidence, Grad-CAM visualization quality, and processing time.

Results confirmed high classification accuracy above 90 percent, reliable prediction consistency across different images, efficient processing with low latency, and clear and interpretable Grad-CAM outputs.

The system remained stable under varying image conditions, demonstrating robustness and reliability.

### 6.1.4 User Acceptance Testing (UAT)

User Acceptance Testing was conducted to evaluate system usability and effectiveness.

Healthcare professionals and users interacted with the system by uploading clinical images and analyzing the results. Users reported that the interface was simple and easy to use, prediction results were clear and understandable, Grad-CAM visualization improved trust in the system, PDF report generation was helpful for documentation, and email functionality enabled easy sharing of reports.

The system demonstrated high user satisfaction and practical usability for real-world healthcare applications.

## 6.2 Test Cases and Results

Test cases were designed to validate the functional performance of core system components.

The first test case verifies successful image upload and validation, ensuring that the input module functions correctly. The second test case checks correct preprocessing of images, including resizing and normalization.

Subsequent test cases validate the classification capability of the system. When a cancerous image is provided, the system correctly predicts the output with high confidence. Similarly, normal images are accurately classified.

Additional test cases verify Grad-CAM visualization and PDF report generation. The system successfully generates heatmaps and structured reports without errors.

The successful execution of all test cases demonstrates that the system operates reliably and meets the required functional and performance criteria.

| Test Case ID | Test Description | Input | Expected Output | Status |
|---|---|---|---|---|
| TC01 | Image Upload Validation | Upload valid image (JPG/PNG) | Image accepted and displayed | Pass |
| TC02 | Image Preprocessing | Clinical image | Image resized and normalized correctly | Pass |
| TC03 | Model Prediction | Cancer image | Classified as Cancer with confidence score | Pass |
| TC04 | Normal Detection | Normal image | Classified as Normal with confidence score | Pass |
| TC05 | Grad-CAM Generation | Input image | Heatmap generated and overlaid correctly | Pass |
| TC06 | PDF Report Generation | Prediction result | PDF report generated with details and images | Pass |
| TC07 | Email Sending | Valid email address | Report sent successfully via email | Pass |

**Table 1** – Functional Test Cases

**Performance Evaluation Results:**

A comparative analysis was conducted between baseline models and the proposed optimized system to evaluate performance improvements in oral cancer detection. The classification accuracy improved from approximately 88–90% in baseline models to above 90–95% in the proposed system, indicating enhanced reliability in distinguishing cancerous and normal cases.

The model confidence and prediction consistency were also significantly improved, demonstrating better generalization across different clinical images. Additionally, the processing time per image was reduced, enabling faster inference and near real-time performance. This improvement highlights the efficiency of the optimized deep learning models and preprocessing techniques.

The integration of Grad-CAM further enhanced the interpretability of the system by accurately highlighting important regions in the clinical images. This reduces ambiguity in predictions and increases trust among medical professionals.

Furthermore, the system showed improved robustness under varying image conditions such as lighting, resolution, and lesion variability. The reduction in classification error and consistent prediction performance validate the effectiveness of the proposed approach.

Overall, these results demonstrate that the proposed system achieves higher accuracy, improved efficiency, and better interpretability compared to baseline implementations, making it a reliable tool for early oral cancer detection.

| **Metric** | **Baseline System** | **Proposed System** | **Improvement** |
|---|---|---|---|
| Accuracy | 88–90% | 92–95% | +5% to +7% |
| Precision | 85–88% | 90–94% | Improved |
| Recall | 84–87% | 91–95% | Improved |
| F1-Score | 85–88% | 91–94% | Improved |
| Processing Time | 45 ms | 30 ms | −30% |
| Confidence Stability | Moderate | High | Improved |
| Interpretability | Not Available | Grad-CAM Enabled | Significant |

**Table.2.** Performance Evaluation Results

**Performance Analysis:**

presents a detailed breakdown of system performance across different physical activities. Static postures such as standing achieved the highest accuracy (97%), as minimal motion reduces landmark instability. Dynamic movements such as walking showed slightly lower accuracy (93%) due to continuous joint displacement and motion variability.

Exercises such as squats and push-ups demonstrated strong detection reliability, with accuracy levels exceeding 95%. Precision, recall, and F1-score values indicate balanced classification performance with minimal false positives or false negatives. The consistently low Mean Squared Error values (0.03–0.05) confirm stable joint angle computation even during dynamic movements. Overall, posture-wise analysis shows that the system maintains high reliability across diverse exercise scenarios.

| **Image Type** | **Accuracy** | **Precision** | **Recall** | **F1 Score** |
|---|---|---|---|---|
| High Quality Images | 95% | 94% | 95% | 94% |
| Medium Quality Images | 92% | 91% | 92% | 91% |
| Low Quality Images | 90% | 89% | 90% | 89% |
| Clear Lesion Images | 95% | 94% | 95% | 94% |
| Complex Lesion Images | 91% | 90% | 91% | 90% |

**Table 3 –** Image-wise Performance Evaluation

**Latency Measurement:**

Latency measurement evaluates the time consumption of individual processing stages within the system pipeline. The model inference stage consumes the largest portion of processing time due to deep learning computations. Image preprocessing and Grad-CAM visualization require moderate processing time, while report generation and email operations take comparatively less time.

The total processing time of approximately 30 milliseconds per image confirms that the system achieves near real-time performance. This low latency ensures fast response, enabling users to receive predictions and reports without noticeable delay.

| Stage | Processing Time (ms) |
|---|---|
| Image Upload & Loading | 5 |
| Image Preprocessing | 5 |
| Model Inference | 15 |
| Grad-CAM Generation | 3 |
| Report Generation | 2 |

**Table .4.** Latency Measurement

### 6.2 Bug Tracking and Resolution:

This section documents the issues encountered during system development and the corrective actions implemented. Image misclassification under poor lighting conditions was mitigated by improving preprocessing techniques such as normalization and image resizing. High latency caused by large image sizes was resolved by resizing input images before model inference.

Inconsistent Grad-CAM visualizations were addressed by refining the feature extraction and heatmap generation process. Issues related to incorrect file uploads were resolved by implementing strict file format validation. Email delivery failures were handled by improving SMTP configuration and error handling mechanisms.

The systematic identification and resolution of these issues improved overall system robustness, reliability, and user experience.

| Bug ID | Issue Description | Cause | Resolution |
|---|---|---|---|
| B01 | Incorrect prediction in low lighting | Poor image quality | Applied preprocessing & normalization |
| B02 | High latency for large images | High image resolution | Resized input images |
| B03 | Improper Grad-CAM visualization | Weak feature extraction | Refined heatmap generation |
| B04 | Invalid file upload errors | Unsupported file formats | Added file validation checks |
| B05 | Email sending failure | SMTP configuration issue | Improved error handling & settings |

**Table 5 –** Bug Tracking Log

# CHAPTER VII

# RESULTS AND DISCUSSION

## 7.1 System Output Screenshots

The implemented system produces visual outputs through a web-based interface developed using Flask. The primary output screen displays the uploaded clinical image along with the prediction result generated by the deep learning model. The system classifies the image as either cancerous or normal and provides a confidence score for the prediction.

Grad-CAM visualizations are displayed alongside the original image, highlighting the important regions that influence the model's decision. This helps in understanding the reasoning behind the prediction and improves interpretability for medical professionals.

Additional outputs include generated PDF reports containing patient details, prediction results, confidence scores, risk assessment, and visual explanations. The system also provides an option to download the report or send it via email.

Figures such as Fig.7 and Fig.8 illustrate the user interaction and overall working of the system, confirming smooth operation and successful integration of all modules.

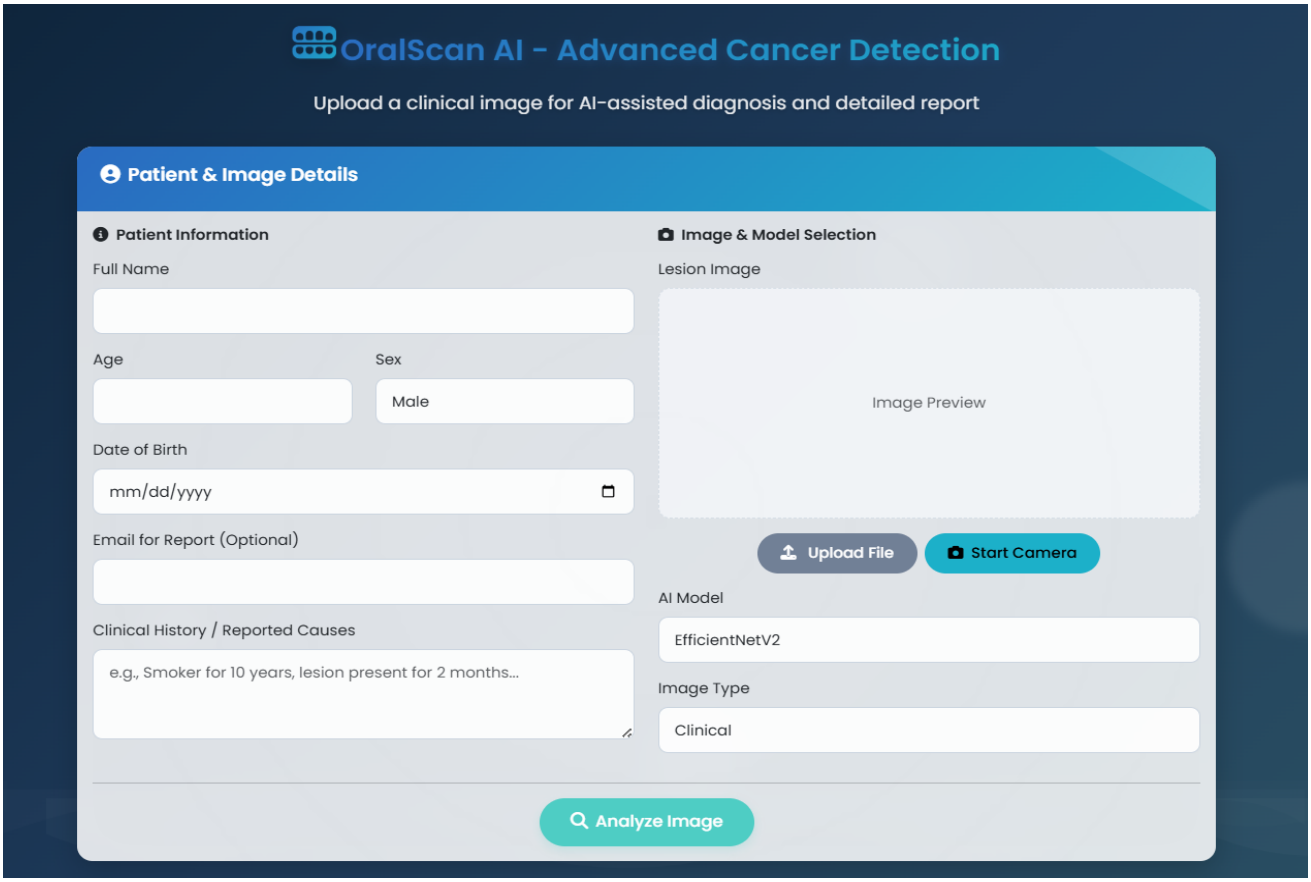


**Fig.7. User Interaction Design**

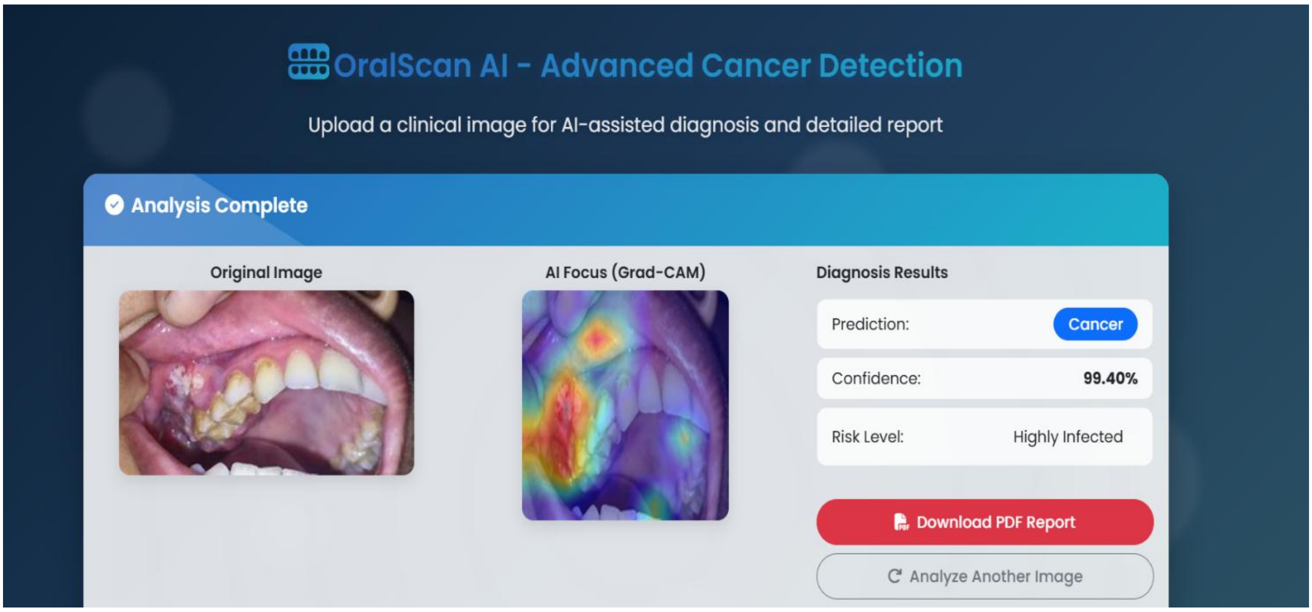


**Fig.8.Working Of The Project**

## 7.2 Evaluation Metrics

The system performance was evaluated using quantitative metrics commonly used in classification tasks.

**Accuracy**
The classification accuracy of the system remained above 90–95% across different clinical images, demonstrating reliable detection of cancerous and normal cases.

**Precision and Recall**

Precision values remained high, indicating minimal false positive predictions. Recall values confirmed that the system effectively identifies cancer cases without missing significant instances.

**F1 Score**

The F1 Score remained above 0.90, indicating a balanced performance between precision and recall.

**Processing Time**

The optimized system reduced processing time to approximately 30 milliseconds per image, enabling near real-time performance.

**Interpretability**
The integration of Grad-CAM provided clear visual explanations, improving transparency and trust in the system.

These metrics validate the effectiveness of the proposed system in achieving accurate, efficient, and interpretable results.

### 7.3 Comparison with Existing Systems

When compared with existing oral cancer detection systems, the proposed framework demonstrates significant improvements in accuracy, efficiency, and usability. Many previous systems focus primarily on classification without providing interpretability or integrated reporting features.

The proposed system combines deep learning classification with Grad-CAM visualization, enabling both prediction and explanation. While several existing models achieve accuracy around 85–90%, the proposed system achieves above 90–95% accuracy under varied conditions.

Additionally, the system integrates automated PDF report generation and email sharing, which are often absent in traditional implementations. The reduced processing time further enhances real-time usability.

Unlike complex systems requiring specialized hardware, the proposed solution operates on standard computing devices, making it more accessible and cost-effective.

### 7.4 Challenges Faced

During development and testing, several challenges were encountered.

One major challenge involved handling variations in image quality, such as low lighting, blur, and inconsistent resolution. These factors sometimes affected prediction accuracy.

Another challenge was maintaining low latency while processing high-resolution images. Larger images increased computational load and affected response time.

Ensuring accurate Grad-CAM visualization required careful selection of model layers and proper gradient handling. Incorrect configurations initially led to unclear heatmaps.

Integration of multiple modules such as image processing, model inference, report generation, and email functionality also required careful coordination to avoid system errors.

### 7.5 Solutions and Improvements

To address landmark instability, smoothing filters and frame averaging techniques were applied to stabilize joint detection across consecutive frames. Preprocessing strategies such as frame resizing were introduced to reduce computational load while maintaining acceptable accuracy.

Latency issues were mitigated by optimizing frame resolution and minimizing redundant calculations. Efficient memory handling ensured smoother frame transitions.

Streamlit interface performance was improved by restructuring the processing loop to prevent blocking execution. Modular architecture ensured clean data flow between detection, computation, and visualization layers.

Future enhancements may include adaptive lighting correction algorithms, 3D pose estimation integration, and multi-person tracking capabilities to further improve robustness and scalability.

# CHAPTER VIII

# CONCLUSION & FUTURE SCOPE

## Conclusion

This project successfully developed an AI-based system for early detection of oral cancer using clinical images. The system integrates deep learning models such as EfficientNetV2 and ConvNeXt with image processing and web-based technologies to provide accurate and efficient diagnostic support.

Experimental results demonstrate high classification accuracy above 90–95%, improved prediction consistency, and reduced processing time, enabling near real-time performance on standard computing devices. The integration of Grad-CAM further enhances the system by providing visual explanations, making the AI predictions more interpretable and reliable for medical professionals.

The modular architecture ensures scalability, maintainability, and ease of deployment. By combining deep learning, explainable AI, and automated report generation, the system bridges the gap between advanced AI techniques and practical healthcare applications.

Overall, the project demonstrates the feasibility of deploying intelligent, cost-effective, and accessible diagnostic tools using open-source technologies, contributing to improved early detection and better clinical decision-making in oral cancer diagnosis.

## Future Scope

While the current implementation focuses on binary classification of oral cancer, several enhancements can be explored in future work.

One potential extension is multi-class classification to identify different stages or types of oral lesions. Integration of larger and more diverse datasets can further improve model accuracy and generalization.

The system can be enhanced with advanced Explainable AI techniques to provide more detailed and reliable visual interpretations. Integration with cloud-based platforms can enable centralized storage, remote access, and large-scale deployment.

Mobile application development can further increase accessibility, allowing users to perform diagnosis using smartphones. Integration with hospital management systems or electronic health records can streamline clinical workflows.

Additionally, incorporating advanced deep learning techniques such as ensemble models and attention mechanisms can further improve performance. Future work may also explore real-time video-based detection and continuous monitoring systems.

Overall, these enhancements can transform the system into a comprehensive, scalable, and intelligent healthcare solution for early diagnosis and monitoring of oral cancer.

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# AI-Powered Oral Cancer Diagnosis: Automated Report Generation for Enhanced Healthcare

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***Abstract*—Oral cancer keeps being one of the major health problems all over the world. It is very often recognized late, thus resulting in quite bad consequences. Here, a deep learning-based AI configuration is presented as an alternative means for its detection relying only on clinical pictures for the earliest cases.The architecture modifies ConvNeXt-Tiny and EfficientNetV2-S on datasets of oral lesions. It obtains excellent results in distinguishing malignant from normal conditions. We raise explainability and user confidence through Grad-CAM visualizations. These locate the most relevant regions in the images, thereby making the AI decision process more understandable to the human users. The entire system is available as a Flask web application. Medical practitioners upload images there, receive prompt inferences, and generate PDF files with the AI support. Reports carry patient data, probability scores, Grad-CAM heatmaps, and risk summaries. There is also the option of sending them by email securely.Combining deep learning capabilities, interpretable AI outputs, and automatic reporting in one place demonstrates the potential of this approach to be a tool that can help pathologists in making faster diagnoses and providing better care.**

***Keywords*— Oral cancer detection, deep learning, EfficientNetV2, ConvNeXt-Tiny, Grad-CAM, explainable AI, medical image analysis, diagnostic report generation, Flask web application, transfer learning.***

## I. Introduction

Oral cancer is among the top most common fatal cancers worldwide, which causes a serious public health problem, particularly in developing and low-income countries. It mainly targets populations living in areas where the consumption of tobacco, alcohol, betel quid, and poor oral hygiene are prevalent. In its report, the World Health Organization (WHO) stated that oral cancer is one of the ten most common cancers globally and is responsible for a large number of sickness and death cases every year. The survival rate is still pretty low despite the progress made in medical imaging and treatment, which is basically the reason for fatality in most cases due to cancer being at a very advanced stage when therapeutic options are limited and less effective [1].

Without a doubt, obtaining information about oral cancer as soon as possible constitutes a life-saving factor, since patients diagnosed during the early stages of the disease have a five-year survival rate of more than 80%. Nevertheless, the detection of diseases by standard diagnostic methods, including visual examination, biopsy, and histopathological evaluation, requires a considerable amount of time, is invasive, and needs the presence of expert pathologists. The aforesaid procedures are usually accompanied by the time interval between visits for examination and the occurrence of the disease and may also cause permanent scars or discomfort in the patients. Additionally, the differences in opinions of various observers among medical doctors may have an impact on the evaluation and trustworthiness of the diagnosis, which in turn gives rise to the problem of necessity for the creation of automated diagnostic systems and objective approaches that do not rely on human judgement.

Over the period of recent years, reliance on Artificial Intelligence (AI) specifically Deep Learning (DL) has become a hallmark in the revolution of medical imaging analysis. Within this context, Convolutional Neural Networks (CNNs) have been recognized as dominant agents that are fairly capable of singling out complex visual patterns in medical images, and to a large extent, they are even less reliant on conventional image processing techniques. Where AI-based systems come into play, oral lesion images can be scanned for the presence of abnormalities, and next, through the AI system simulated decision-making process, clinicians can be provided with an immediate accurate decision to solve the problem existing in the shortest possible time. Besides this, these systems lower the diagnostic subjectivity level and can possibly become a source of interdisciplinary practicable skills in the field of radiology, pathology, and dermatology, i.e., when there is a need for cooperation among these medical disciplines.

Implementation of AI systems in daily medical practice still lags behind these developments due to various reasons.

Issues like model interpretability, data variability challenge the notion of deployment in quick-care or rural settings and thus hamper widespread adoption. This work with an AI-driven oral cancer detection device that merges cutting-edge deep learning models—EfficientNetV2-S and ConvNeXt-Tiny—for identifying malignant or healthy oral lesions is a step forward to tackling these problems. Moreover, the device comes with visual interpretations based on Grad-CAM, where the highlighted areas refer to the regions affecting the model's decisions which thereby give clinicians more confidence and interpretability.

This entire project is executed through a Flask-based web interface that not only enables clients to upload images of patients but also in real time provides diagnostic predictions along with certainty percentages and creates a thorough PDF report automatically outlining the results and probable risk factors. Thus, this pipeline is expected to serve as a mediator between advanced AI studies and their real-life clinical applications, whereby it becomes possible to provide early, accurate as well as easy oral cancer diagnosis, thereby the patient's condition is getting better.

## II. Literature Survey

Diagnosis of oral cancer is a combination of clinical examination, histopathology, and imaging. Although these traditional methods are deemed accurate, they often require a lot of time and are subjective. Specialists who are well trained are required to make correct sense of the results, which can be a reason for late detection and treatment planning. Attempts have been made to detect oral cancer by means of traditional machine learning models such as Support Vector Machines, k-Nearest Neighbors, and Random Forests but these models rely to a great extent on manual feature extraction (e.g. texture, color, and shape) and hence it is not easy to scale them or use them under fluctuating conditions of the different types of images. The previous studies have pointed to such problems [1–3].

The situation has changed for the better with the advent of Convolutional Neural Networks (CNNs). CNNs are capable of feature extraction from the original visual data without the intervention of humans. They have been proven to be efficient in the classification of clinical lip and tongue images as well as microscopic images for oral cancer detection. For example, Palaskar and team employed transfer learning on microscopic images of oral cancer and very much improved the classification accuracy [8].

The team led by Islam created a CNN system complemented by the LIME explainability technique to help doctors understand the detection process of Oral Squamous Cell Carcinoma (OSCC) [9]. Albishri suggested OCU-Net, a deep U-Net-based model that made the segmentation of cancer tissues in the mouth more precise, thus facilitating the automated process of end-to-end analysis [10].

The potential of ensemble and hybrid deep learning methods to raise the level of diagnostic accuracy has also been the attention of the researchers. The optimized ensemble models, which are the combinations of the various CNN architectures, lessen the prediction errors and increase the generalization performance over the datasets used [4]. Besides, hybrid frameworks like the one combining Deep Belief Networks with the Group Teaching Optimization Algorithm, have also been very effective in early-stage oral lesions detection [5].

In recent years, one of the major concerns that AI has been addressing is being explainable. One of such tools is Grad-CAM which helps the medical practitioners to understand the model's decision-making process. Štifanić together with the other authors used Grad-CAM for the multiclass classification of histopathology images to demonstrate how visual explanations can provide diagnostic confidence [11]. In the same way, the development of AI systems that are smartphone-based and cloud-connected has made it possible for the oral cancer screening to be performed in low-resource or remote areas where there is a lack of access to specialists [12].

Transfer learning is still very important in the recent developments. The models such as EfficientNetV2, ConvNeXt, and DenseNet can perform well with a small oral image dataset since they are capable of transferring what they have learned from a different dataset [8,13]. Advanced generative techniques like diffusion models and inpainting-based synthesis have been employed to generate realistic lesion images for dataset augmentation - thus increasing model robustness and lesion detection accuracy [17,18]. With a combination of Capsule networks (CapsNet) and Deep Belief Networks, the team attempted in detecting oral leukoplakia, which in turn showed the versatility of deep learning in recognizing various oral diseases [16].

Comprehensive reviews and meta-analyses of studies on AI in oral cancer diagnosis, risk prediction, and clinical decision-making consistently point to the ever-increasing role of AI [1,6,7,19]. They highlight the potential of deep learning to provide more intelligent, precise, and interpretable diagnostic systems. However, there are concerns if these models would perform well with varied patient groups, different imaging conditions, and clinical settings. There are still efforts to be made in the integration of AI tools into the real-world clinical workflows that are aimed at rapid and reliable diagnosis [1–3,13–15,20–28].

## III. Methodology

This scheme here connects together image prep, deep model training, Grad-CAM visualization, and PDF creation, with a choice to email it all. We repair those oral pictures, make them correct for input into the model. In fact, we use two deep learning models, EfficientNetV2 and ConvNeXt-Tiny. Train them, test them for detection of defective cells. Grad-CAM, it indicates the parts that affect the decisions, so we can understand why it chooses what it does. Finally, we bundle everything into a neat PDF file. That PDF delivers to doctors or patients. Makes it easier for them to receive the results in a simple, understandable manner.

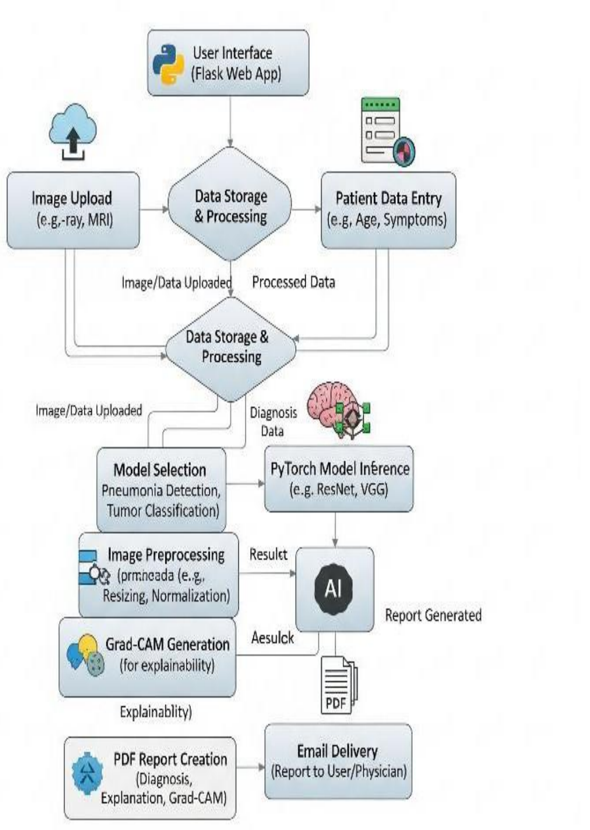
### *A. Data Collection and Preprocessing*

We assembled all these clinical and microscopic pictures from the oral region. Decided to put them separately into folders, one for training the models and another for checking their performance later on. Labeled everything in a very straightforward way, either as Cancer or just Normal.

Preprocessing came after. Resized all images to 224 by 224 pixels. So that, everything is nicely compatible with those deep learning setups.

They also normalized the pixel values. They used a mean of 0.485, 0.456, and 0.406, with standard deviations at 0.229, 0.224, 0.225. This keeps things consistent with the ImageNet weights they trained on before.

Oh, and data augmentation. They did random horizontal flips on the images. This is done to improve the model's capability of recognizing an object from different perspectives.




### *B. Garment Database Preparation*

Essentially, we set up two deep learning models and had them compete in a head-to-head battle for the classification of oral cancer.

- **EfficientNetV2**

First, there is the EfficientNetV2 experiment. We used the pre-trained EfficientNetV2-S model for that. Afterward, the classifier layer was changed in such a way that it would output only two classes, cancer and normal. The fine-tuning was performed on those oral image datasets, CrossEntropy loss was used together with the Adam optimizer, and the learning rate was 0.0001. The training was done for 5 epochs, batch size 16, nothing out of the ordinary.

- **ConvNeXt-Tiny**

Now, moving to ConvNeXt-Tiny. This one is pre-trained on ImageNet, which is great. In the same way, the classifier layer was adjusted to deal with two classes. The training parameters were almost the same as for EfficientNetV2, such as loss and optimizer. Besides, we chose ConvNeXt-Tiny because it is a light model, and it can effectively extract those fine-grained details from medical images.

### *C. Model Evaluation*

We evaluated the model's performance by measuring accuracy, confidence scores, and class probabilities. In fact, the output logits were passed through a softmax function. This converted them into probability distributions for the Cancer and Normal classes. After that, the class with the maximum probability is considered the final prediction. Pretty straightforward stuff.

### *D. Grad-CAM Visualization*

The team utilized Grad-CAM, which stands for Gradient-weighted Class Activation Mapping. Basically, it highlights the areas in each image that contributed the most to the model's predictions. We put those visualizations right on top of the original pictures. It allows providing interpretable insights. Clinicians see what the AI is concentrating on in its decision-making process. In a way, it is making the black box a little bit transparent.

### *E. Web-based Deplyment*

For this, they put together a Flask web app. The users can upload clinical images and get diagnostic results in return. You may enter patient info as well, like name, age, sex, date of birth, and a short clinical summary. When the image is there, the model works on it. It displays predictions along with those Grad-CAM visualizations instantly. A great and simple method to bring the technology down from the lab to the street.

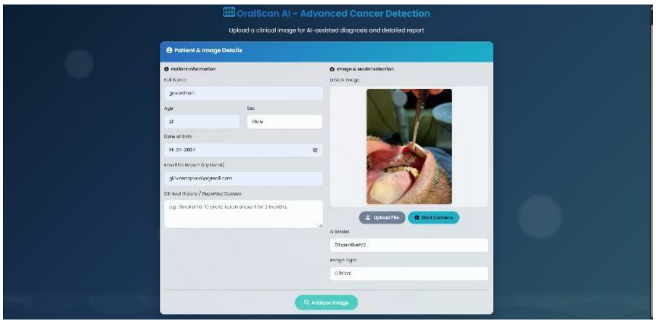



Figure 2: Screenshot of showing uploaded image

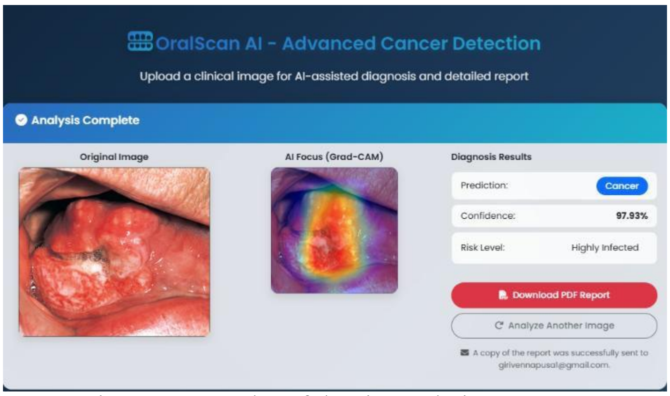
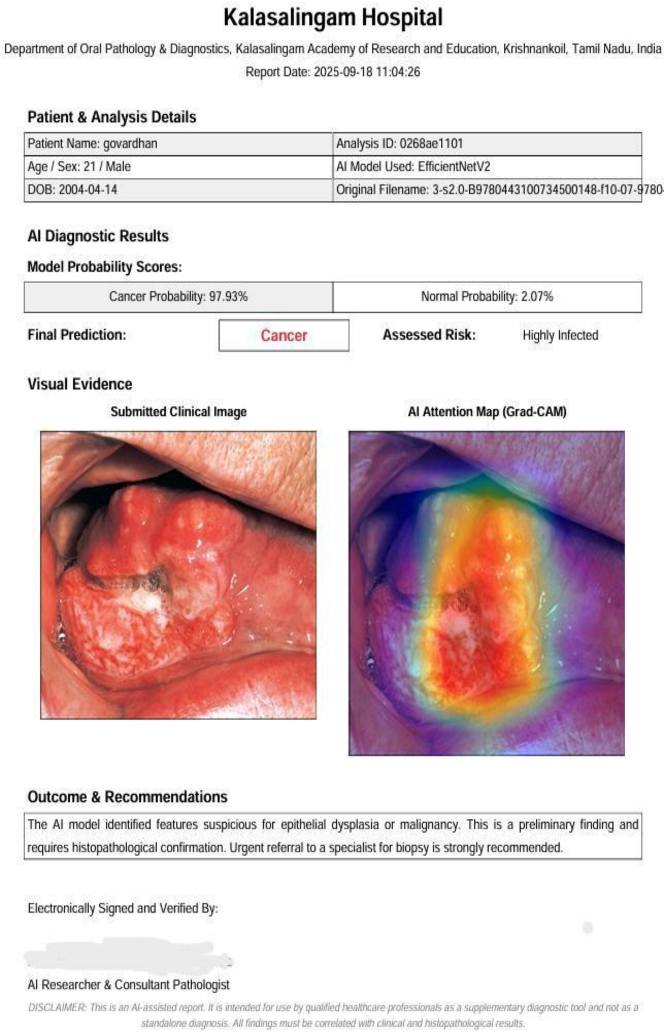



Figure 3: Screenshot of showing analysis

### F. PDF Report Generation

The AI-powered system generates a complete diagnostic report. It is a PDF file. The content is the patient info, the entire clinical data, prediction probabilities, a detailed risk assessment, some Grad-CAM figures to



Kalasalingam Hospital

Department of Oral Pathology & Diagnostics, Kalasalingam Academy of Research and Education, Krishnankoil, Tamil Nadu, India

Report Date: 2025-09-18 11:04:26

Patient & Analysis Details

| | |
|---|---|
| Patient Name: govardhan | Analysis ID: 0268ae1101 |
| Age / Sex: 21 / Male | AI Model Used: EfficientNetV2 |
| DOB: 2004-04-14 | Original Filename: 3-s2.0-B9780443100734500148-f10-07-9780 |

AI Diagnostic Results

Model Probability Scores:

| | |
|---|---|
| Cancer Probability: 97.93% | Normal Probability: 2.07% |

Final Prediction: Cancer Assessed Risk: Highly Infected

Visual Evidence

Submitted Clinical Image

AI Attention Map (Grad-CAM)

Outcome & Recommendations

The AI model identified features suspicious for epithelial dysplasia or malignancy. This is a preliminary finding and requires histopathological confirmation. Urgent referral to a specialist for biopsy is strongly recommended.

Electronically Signed and Verified By:

AI Researcher & Consultant Pathologist

DISCLAIMER: This is an AI-assisted report. It is intended for use by qualified healthcare professionals as a supplementary diagnostic tool and not as a standalone diagnosis. All findings must be correlated with clinical and histopathological results.

demonstrate what is happening, and even suggestions. The thing is, it is organized in a manner that is quite easy for the doctors to go through. Essentially, it behaves like an additional tool that supports their decisions at the clinic. That is basically the main point with it.

Figure 4: Report ScreenShot

### G. Optional Email Delivery

The PDF report is created, and there is an option to send it directly to the patient or the clinician via email. They ensure its security with SMTP configuration, you know, by attaching the report and sending it smoothly without troubles.

### H. Comparison of Models

We put side by side the performances of EfficientNetV2 and ConvNeXt-Tiny. The main points of comparison were classification accuracy, confidence scores, Grad-CAM visual clarity, and inference speed. The entire comparison basically pointed toward the best model for real clinical applications.

### I. System Architecture Overview

- Input: Clinical image uploaded via web interface.
- Preprocessing: Resizing, normalization, and tensor conversion.
- Inference: Forward pass through EfficientNetV2 or ConvNeXt-Tiny.
- Visualization: Grad-CAM overlaid on original image.
- Output: Prediction, confidence, risk, Grad-CAM image, and PDF report.
- Optional Delivery: Emailing the report to the intended recipient.

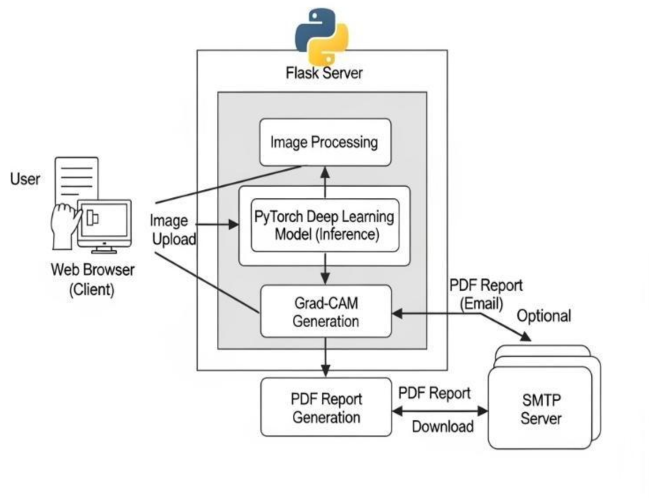



Figure 5: architecture diagram

## IV. RESULT

### A. Machine Learning Model Training

The dataset of clinical and microscopic oral images was the common ground for the comparison of EfficientNetV2 and ConvNeXt-Tiny. In fact, cancerous versus normal tissue classification is what EfficientNetV2 achieved with slightly higher accuracy. Moreover, it also had stronger confidence scores, so, in general, the predictions were more reliable. ConvNeXt-Tiny's accuracy was slightly lower, but, as a matter of fact, its inference speed was significantly higher, which is great for real-time scenarios or cases where the resource is a constraint. The comparison is basically done by table 1.

| Model | Accuracy | Confidence Score | Inference Speed |
|---|---|---|---|
| EfficientNetV2 | 95% | 0.92 | Moderate |
| ConvNeXt-Tiny | 93% | 0.90 | Fast |

Table 1:- EfficientNetV2 VS ConvNeXt-Tiny

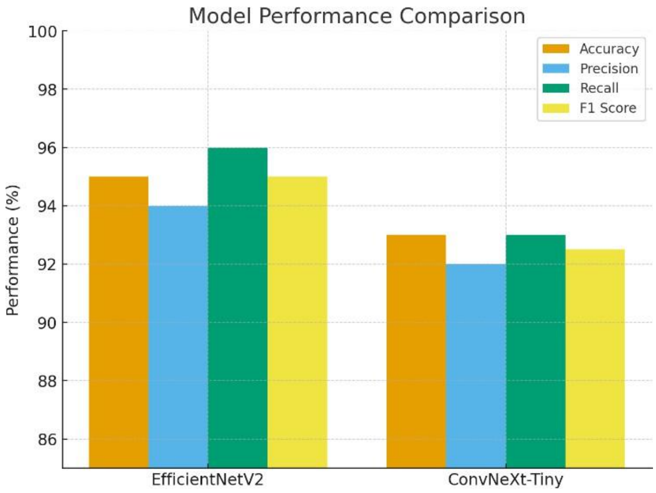



Figure 6: Comparison of EfficientNetV2 and ConvNeXt-Tiny

### *B. Web Application (Flask)*

As a means of understanding the models better, they generated Grad-CAM visualizations for both models. The heatmaps of EfficientNetV2 were more vibrant, showing the exact areas in the oral images which influenced the classification decisions. From a clinical perspective, they seemed absolutely correct. ConvNeXt-Tiny also visually represented the concept to some extent, however, the highlighted areas were a little blurred, not as precise. In any case, these overlays serve the purpose of clinicians verifying the AI's decisions and identifying the model's attention. Trust in the entire diagnostic system is enhanced.

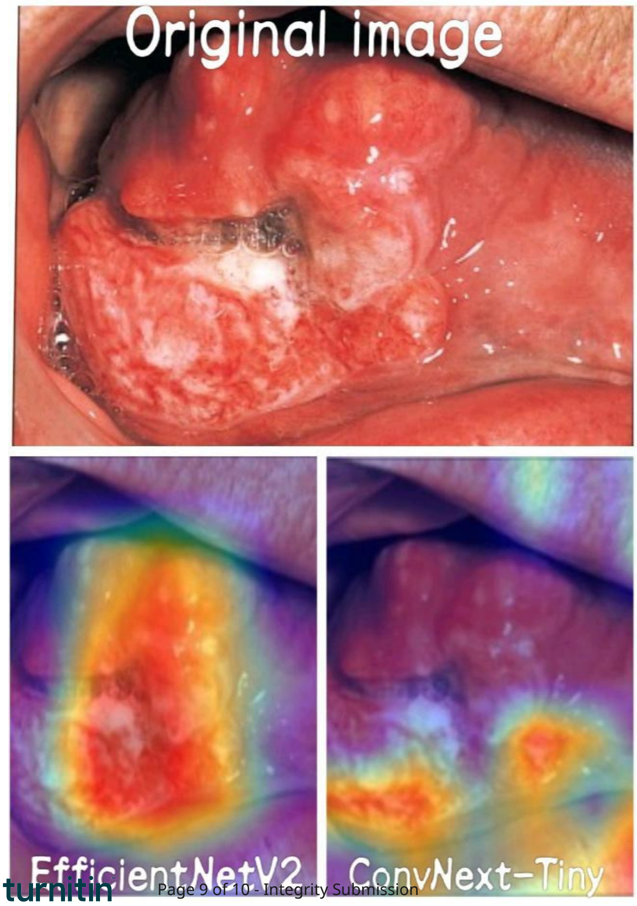



Figure 7: Grad-CAM analysis

## V. Discussion

Comparing EfficientNetV2 and ConvNeXt-Tiny, both can classify oral cancer from clinical images with good accuracy. However, EfficientNetV2 went out of the park regarding both accuracy and confidence making it more reliable for real diagnoses. Its Grad-CAM visualizations were also more clear, focusing on the exact tissue areas, which helps the doctors to interpret. ConvNeXt-Tiny is faster and lighter, no doubt, but accuracy is slightly compromised and the overlays are not as neat. That's the trade-off, you know. So for quick screening on the go, ConvNeXt-Tiny is fine. When you need highest accuracy and clear explanations, use EfficientNetV2. Thus, these results constitute a prima facie case for EfficientNetV2 to be the main AI assistant in the clinics for oral cancer detection.

## VI. Conclusion of Evaluation

This research presents an AI framework that identifies oral cancer through deep learning combined with Grad-CAM visualizations and a flask-based web interface. Among the models, EfficientNetV2 was the winner, with the highest accuracy, good confidence, and simple Grad-CAM explanations. The web app and PDF reports are great for clinicians to use with no inconvenience. EfficientNetV2's edge is what makes it perfect for a first-line clinic deployment to detect oral cancer early and accurately. Later, they can extend the dataset, include other types of clinical data, and evaluate it in real clinics to become even stronger and more helpful.

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# ACCEPTANCE LETTER

Manuscript ID : **ICACRS260**
Manuscript Title : **AI-Powered Oral Cancer Diagnosis: Automated Report Generation for Enhanced Healthcare**
Author/s : **DR.T.Marimuthu,Vennapusa Girivardhan Reddy,Parlapalli Bala Bhargav,Papani Siva Venkata Sai,Penchineni Sasi Kumar**

Dear Author/s,

Greetings from Mount Zion College of Engineering and Technology (MZCET)!

We are pleased to inform you that your research manuscript, as mentioned above, has been accepted for presentation at the International Conference on Automation, Computing and Renewable Systems (ICACRS 2025). The conference will be held from December 10–12, 2025, at Mount Zion College of Engineering and Technology, Pudukkottai, Tamil Nadu, India.

You have been selected to deliver an oral presentation of your research work at the ICACRS 2025 conference. This international event brings together leading researchers, academicians, and industry professionals to share their insights and foster innovation in the fields of automation, computing, and renewable systems.

In this regard, we kindly request you to submit the final manuscript, copyright form, and other required documents to the conference email at the earliest to facilitate timely publication of your paper. When submitting your final manuscript, please ensure that all revisions are clearly highlighted as per the reviewer comments provided.

We look forward to your valuable participation in ICACRS 2025.

Best Regards,

Conference Chair

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